

Linear models

Francisco Rodríguez-Sánchez

<https://frodriguezsanchez.net>

Example dataset: forest trees

- Download [this dataset](#) (or the entire [zip file](#))

```
trees <- read.csv("data/trees.csv")  
head(trees)
```

	site	dbh	height	sex	dead
1	4	29.68	36.1	male	0
2	5	33.29	42.3	male	0
3	2	28.03	41.9	female	0
4	5	39.86	46.5	female	0
5	1	47.94	43.9	female	0
6	1	10.82	26.2	male	0

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- Import:

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5	1	47.94	43.9	female	0
6	1	10.82	26.2	male	0

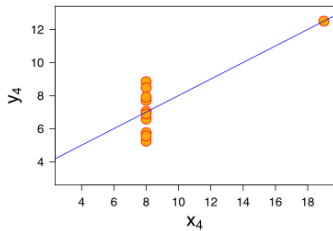
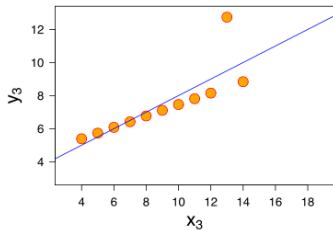
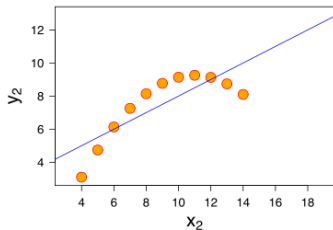
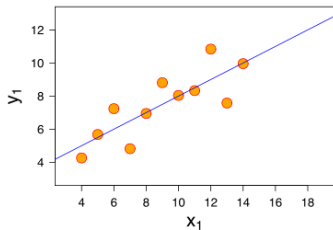
- What is the relationship between DBH and height?

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- Do taller trees have bigger trunks?

- What is the relationship between DBH and height?
- Do taller trees have bigger trunks?
- Can we predict height from DBH? How well?

Always plot your data first!

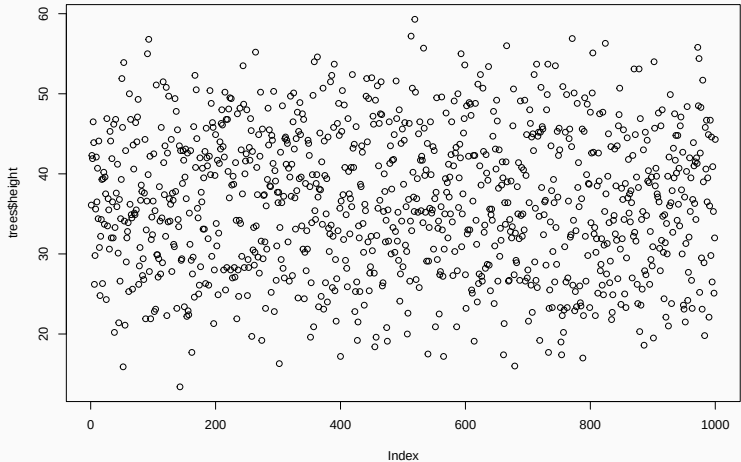
Always plot your data first!



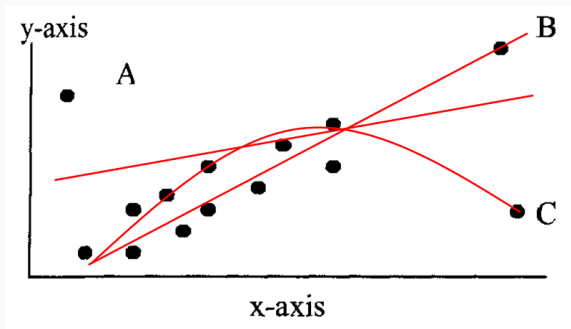
Exploratory Data Analysis (EDA)

Outliers

```
plot(trees$height)
```



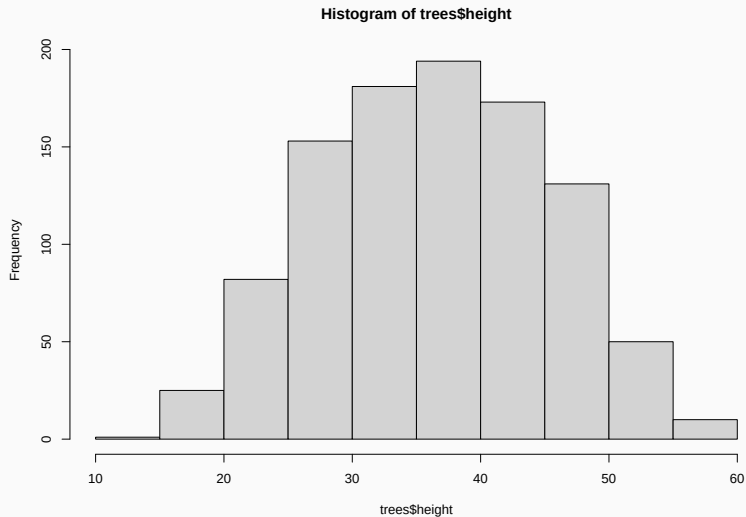
Outliers impact on regression



See <http://rpsychologist.com/d3/correlation/>

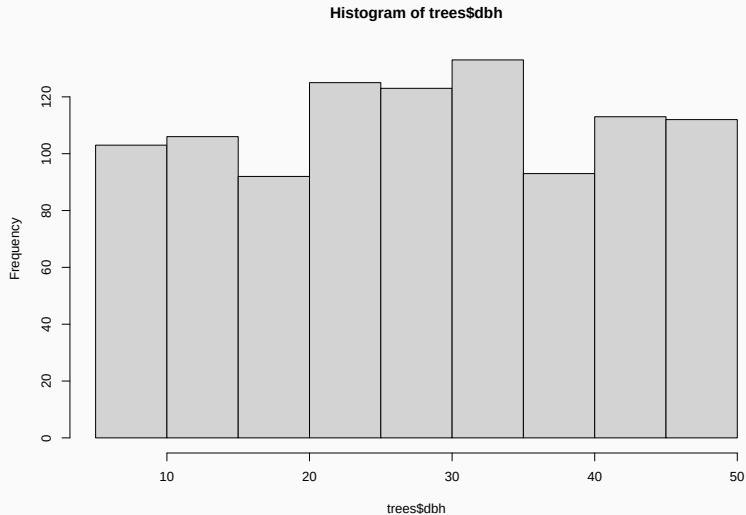
Histogram of response variable

```
hist(trees$height)
```



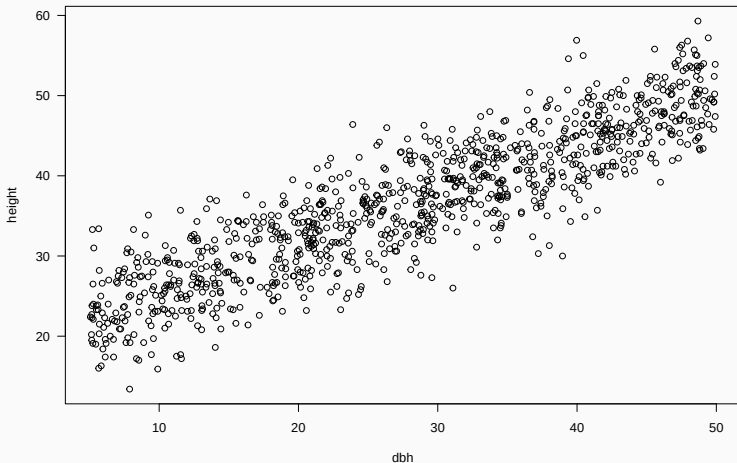
Histogram of predictor variable

```
hist(trees$dbh)
```



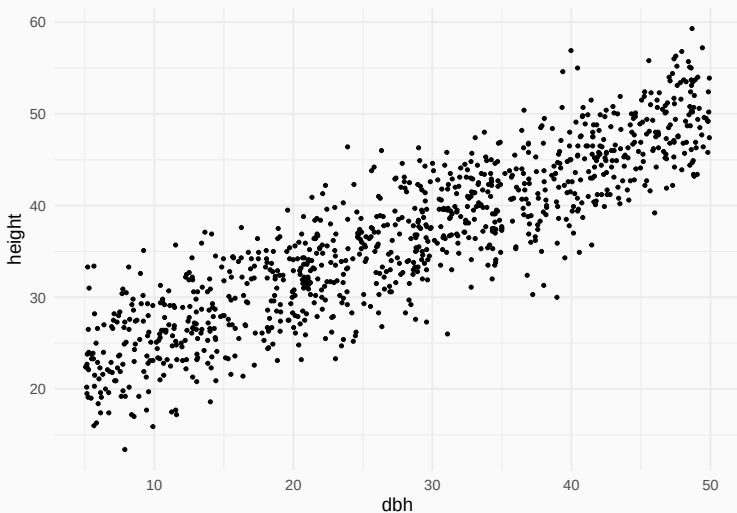
Scatterplot

```
plot(height ~ dbh, data = trees, las = 1)
```



Scatterplot

```
ggplot(trees) +  
  geom_point(aes(x = dbh, y = height))
```



Model fitting

Now fit model

Hint: `lm`

Now fit model

Hint: `lm`

```
m1 <- lm(height ~ dbh, data = trees)
```

which corresponds to

$$\begin{aligned} \text{Height}_i &= a + b \cdot \text{DBH}_i + \varepsilon_i \\ \varepsilon_i &\sim N(0, \sigma^2) \end{aligned}$$

Package `equatiomatic` returns model structure

```
library("equatiomatic")  
m1 <- lm(height ~ dbh, data = trees)  
equatiomatic::extract_eq(m1)
```

$$\text{height} = \alpha + \beta_1(\text{dbh}) + \epsilon \quad (1)$$

```
equatiomatic::extract_eq(m1, use_coefs = TRUE)
```

$$\widehat{\text{height}} = 19.34 + 0.62(\text{dbh}) \quad (2)$$

To preview LaTeX:

```
library(texPreview)  
tex_preview(equatiomatic::extract_eq(m1))
```

Model interpretation

What does this mean?

```
summary(m1)
```

Call:

```
lm(formula = height ~ dbh, data = trees)
```

Residuals:

Min	1Q	Median	3Q	Max
-13.3270	-2.8978	0.1057	2.7924	12.9511

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	19.33920	0.31064	62.26	<2e-16 ***
dbh	0.61570	0.01013	60.79	<2e-16 ***

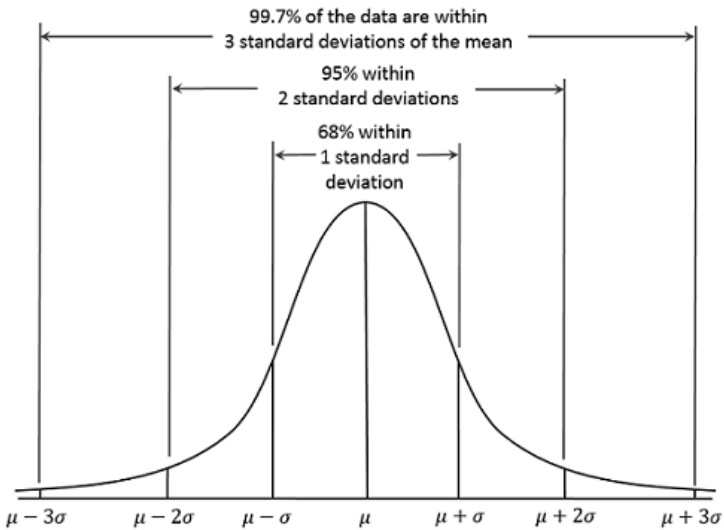
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.093 on 998 degrees of freedom

Multiple R-squared: 0.7874, Adjusted R-squared: 0.7871

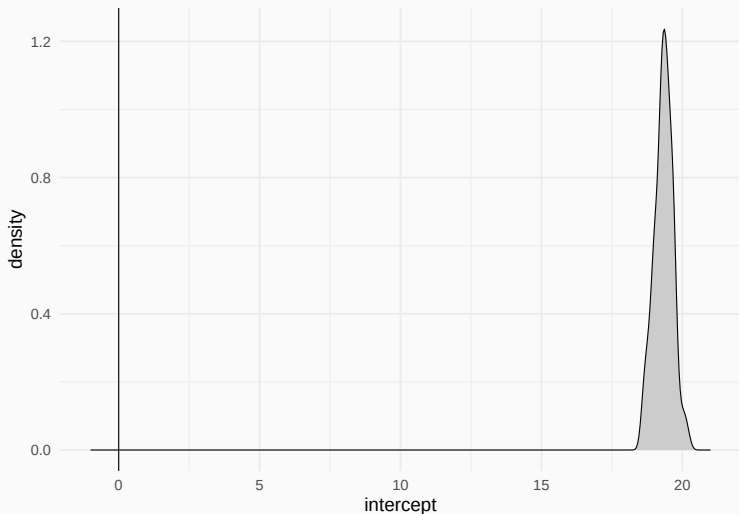
F-statistic: 3695 on 1 and 998 DF, p-value: < 2.2e-16

Remember that in a Normal distribution



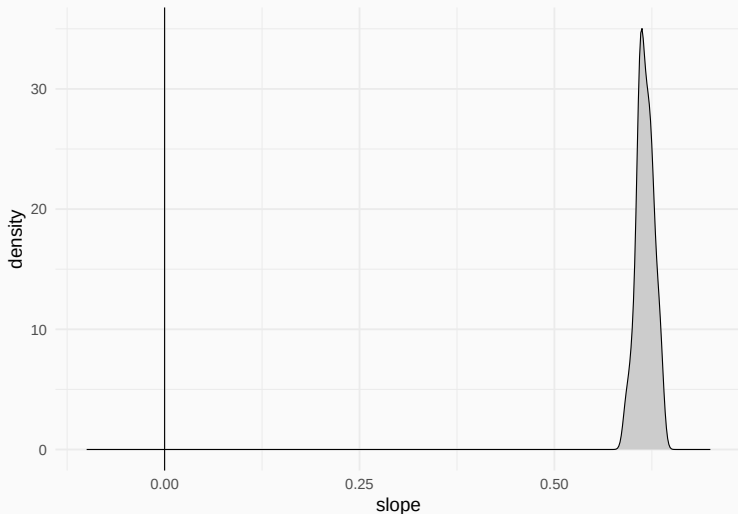
Estimated distribution of the intercept parameter

Parameter	Coefficient	SE	95% CI	t(998)	p
(Intercept)	19.34	0.31	[18.73, 19.95]	62.26	< .001

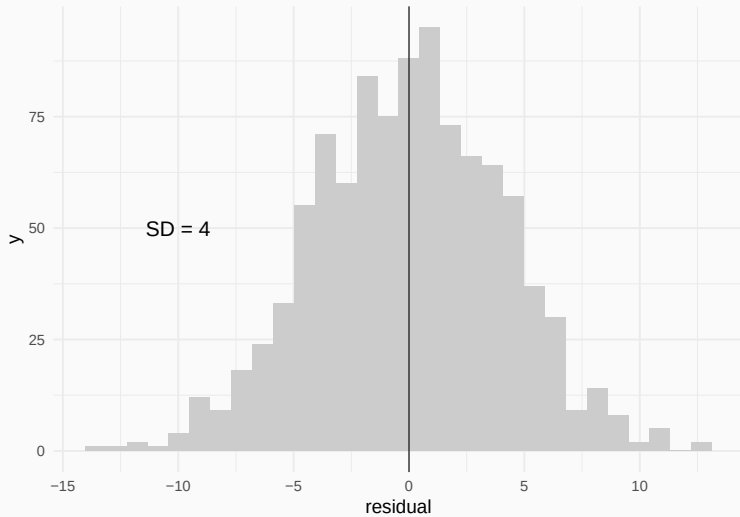


Estimated distribution of the slope parameter

Parameter	Coefficient	SE	95% CI	t(998)	p
dbh	0.62	0.01	[0.60, 0.64]	60.79	< .001



Distribution of residuals



$$DF = n - p$$

n = sample size

p = number of estimated parameters

Proportion of 'explained' variance

$$R^2 = 1 - \frac{\text{Residual Variation}}{\text{Total Variation}}$$

Accounts for model complexity
(number of parameters)

$$R_{adj}^2 = 1 - (1 - R^2) \frac{n-1}{n-p-1}$$

<https://pollev.com/franciscorod726>

```
coef(m1)
```

(Intercept)	dbh
19.3391968	0.6157036

Confidence intervals for parameters

```
confint(m1)
```

	2.5 %	97.5 %
(Intercept)	18.7296053	19.948788
dbh	0.5958282	0.635579

Tidy up model coefficients with broom

```
library("broom")  
tidy(m1)
```

```
# A tibble: 2 x 5  
  term          estimate std.error statistic p.value  
  <chr>          <dbl>    <dbl>    <dbl>    <dbl>  
1 (Intercept)    19.3      0.311     62.3      0  
2 dbh             0.616     0.0101    60.8      0
```

```
glance(m1)
```

```
# A tibble: 1 x 12  
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC  
    <dbl>         <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>  
1    0.787         0.787  4.09    3695.      0     1 -2827. 5660. 5675.  
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

<https://broom.tidymodels.org/>

Retrieving model parameters with `parameters` package

```
library("parameters")  
parameters(m1)
```

Parameter	Coefficient	SE	95% CI	t(998)	p
(Intercept)	19.34	0.31	[18.73, 19.95]	62.26	< .001
dbh	0.62	0.01	[0.60, 0.64]	60.79	< .001

<https://easystats.github.io/parameters/>

Communicating results



- “Never conclude there is ‘no difference’ or ‘no association’ just because $p > 0.05$ or CI includes zero”



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- Estimate and communicate **effect sizes and their uncertainty**



- “Never conclude there is ‘no difference’ or ‘no association’ just because $p > 0.05$ or CI includes zero”
- Estimate and communicate effect sizes and their uncertainty
- <https://doi.org/10.1038/d41586-019-00857-9>

- We found a **significant relationship** between DBH and Height ($p < 0.05$).

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- (add p-value if you wish)

Models that describe themselves

```
library("report")  
report(m1)
```

We fitted a linear model (estimated using OLS) to predict height with dbh (formula: `height ~ dbh`). The model explains a statistically significant and substantial proportion of variance ($R^2 = 0.79$, $F(1, 998) = 3695.40$, $p < .001$, adj. $R^2 = 0.79$). The model's intercept, corresponding to $\text{dbh} = 0$, is at 19.34 (95% CI [18.73, 19.95], $t(998) = 62.26$, $p < .001$). Within this model:

- The effect of dbh is statistically significant and positive ($\beta = 0.62$, 95% CI [0.60, 0.64], $t(998) = 60.79$, $p < .001$; Std. $\beta = 0.89$, 95% CI [0.86, 0.92])

Standardized parameters were obtained by fitting the model on a standardized version of the dataset. 95% Confidence Intervals (CIs) and p-values were computed using a Wald t-distribution approximation.

<https://easystats.github.io/report/>

Generating table with model results: `report_table`

```
report_table(m1, include_effectsize = FALSE)
```

Parameter	Coefficient	95% CI	t(998)	p	Fit
(Intercept)	19.34	[18.73, 19.95]	62.26	< .001	
dbh	0.62	[0.60, 0.64]	60.79	< .001	
AIC					5660.25
AICc					5660.27
BIC					5674.97
R2					0.79
R2 (adj.)					0.79
Sigma					4.09

Generating table with model results: gtsummary

```
library("gtsummary")  
tbl_regression(m1, intercept = TRUE)
```

Characteristic	Beta	95% CI ¹	p-value
(Intercept)	19	19, 20	<0.001
dbh	0.62	0.60, 0.64	<0.001

¹CI = Confidence Interval

<https://www.danielsjoberg.com/gtsummary>

Generating table with model results: `modelsummary`

```
library("modelsummary")  
modelsummary(m1, output = "markdown") # Word, PDF, PowerPoint, png...
```

	(1)
(Intercept)	19.339
	(0.311)
dbh	0.616
	(0.010)
Num.Obs.	1000
R2	0.787
R2 Adj.	0.787
AIC	5660.3
BIC	5675.0
Log.Lik.	-
	2827125
F	3695.395
RMSE	4.09

Generating table with model results: `modelsummary`

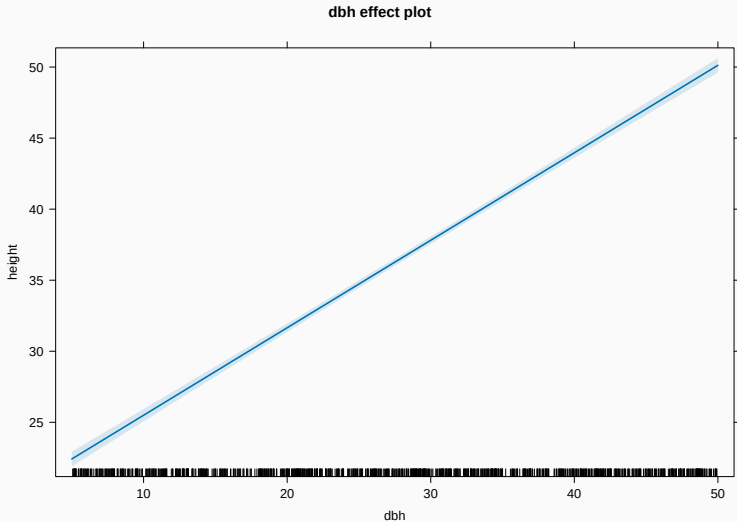
```
modelsummary(m1, fmt = 2,  
  estimate = "{estimate} ({std.error})",  
  statistic = NULL,  
  gof_map = c("nobs", "r.squared", "rmse"),  
  output = "markdown") # Word, PDF, PowerPoint, png...
```

	(1)
(Intercept)	19.34 (0.31)
dbh	0.62 (0.01)
Num.Obs.	1000
R2	0.787
RMSE	4.09

Visualising fitted model

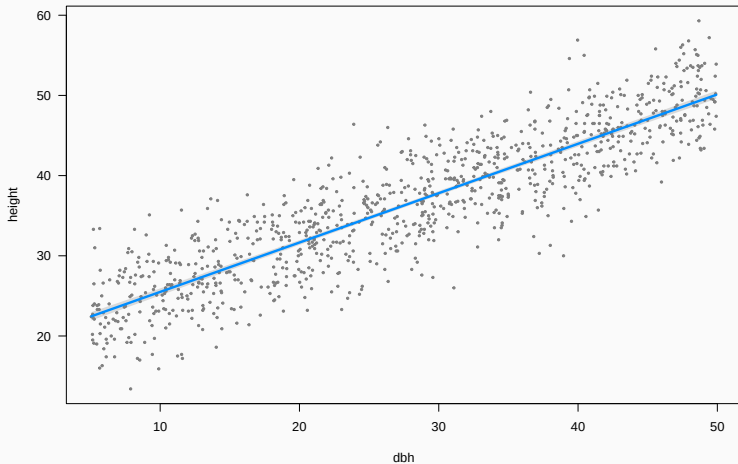
Plot model: effects package

```
library("effects")  
plot(allEffects(m1))
```



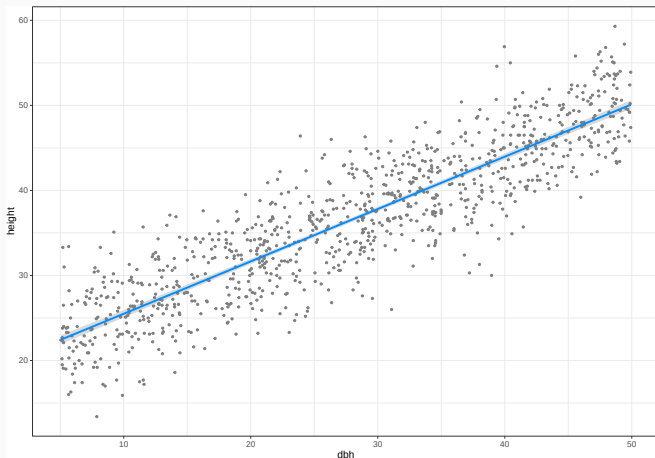
Plot model: visreg

```
library("visreg")  
visreg(m1)
```



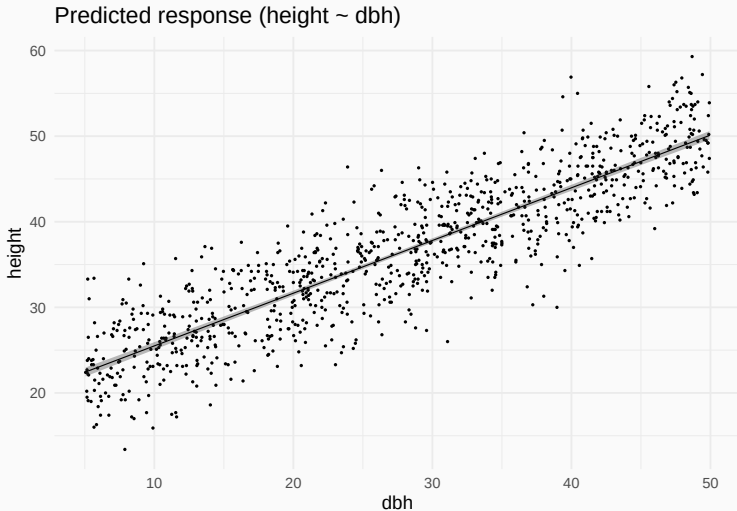
visreg can use ggplot2 too

```
visreg(m1, gg = TRUE) + theme_bw()
```



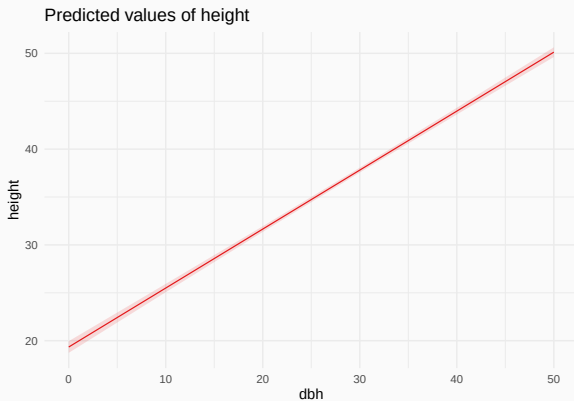
<https://pbreheny.github.io/visreg>

```
library("easystats")  
plot(estimate_expectation(m1))
```



Plot model: sjPlot

```
library("sjPlot")  
plot_model(m1, type = "eff", terms = "dbh")
```



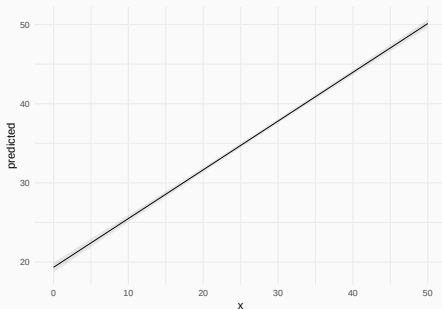
<https://strengjacke.github.io/sjPlot>

```
library("ggeffects")
```

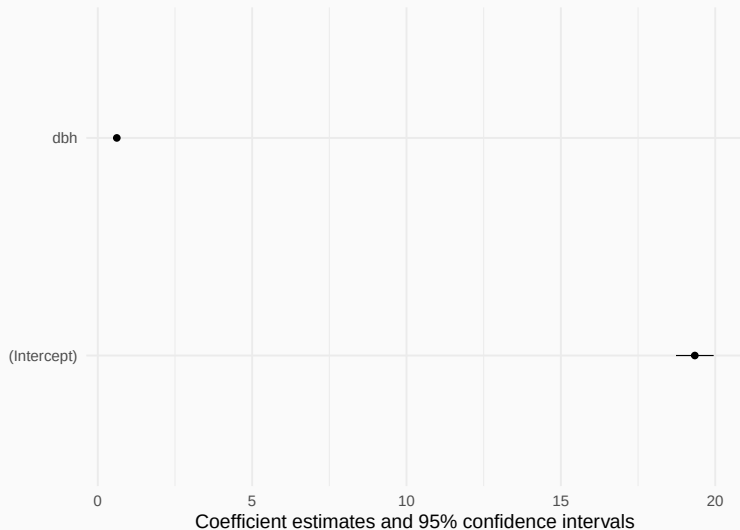
```
mydf <- ggpredict(m1, terms = "dbh")
dplyr::glimpse(mydf, width = 40)
```

```
Rows: 6
Columns: 6
$ x          <dbl> 0, 10, 20, 30, 40, 50
$ predicted  <dbl> 19.33920, 25.49623, ~
$ std.error  <dbl> 0.3106446, 0.2226051~
$ conf.low   <dbl> 18.72961, 25.05941, ~
$ conf.high  <dbl> 19.94879, 25.93306, ~
$ group      <fct> 1, 1, 1, 1, 1, 1
```

```
ggplot(mydf, aes(x, predicted)) +
  geom_line() +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
            alpha = 0.1)
```

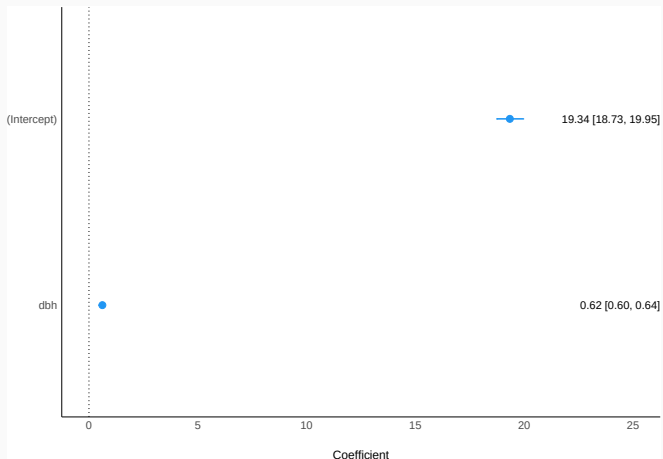


```
modelplot(m1)
```



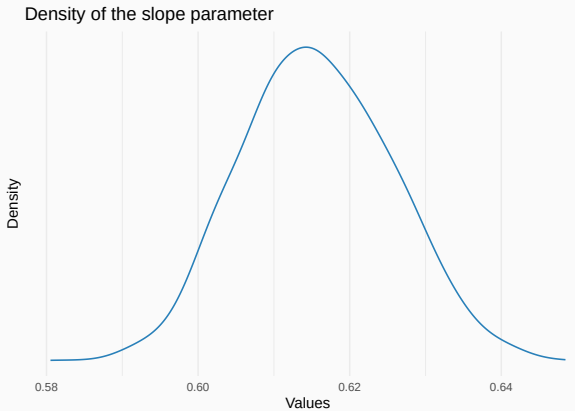
Plot model parameters with easystats (see package)

```
library("easystats")  
plot(parameters(m1), show_intercept = TRUE, show_labels = TRUE)
```



Plot parameters' estimated distribution

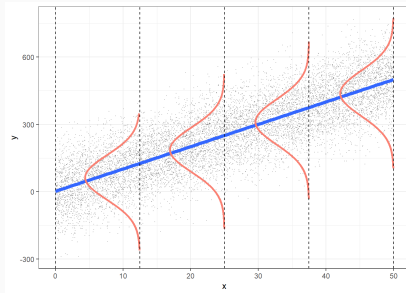
```
plot(simulate_parameters(m1)) +  
  labs(title = "Density of the slope parameter")
```



Model checking

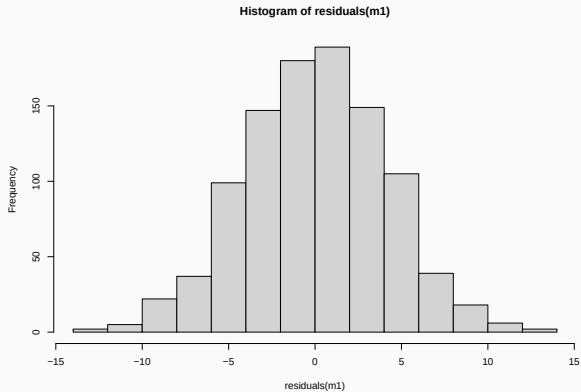
Linear model assumptions

- Linearity (transformations, GAM...)
- Residuals:
 - Independent
 - Equal variance
 - Normal
- Negligible **measurement error** in predictors



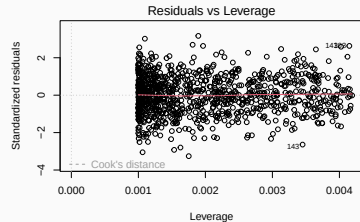
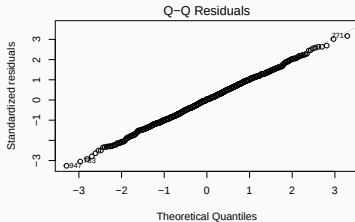
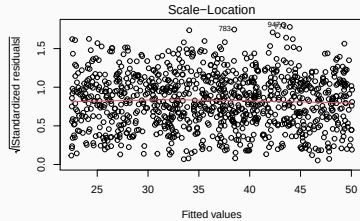
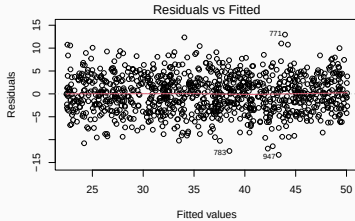
Are residuals normal?

```
hist(residuals(m1))
```



SD = 4.09

Model checking: `plot(model)`

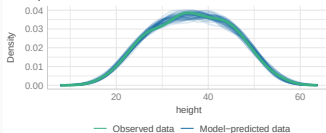


Model checking with performance (easystats)

```
library("easystats")  
check_model(m1)
```

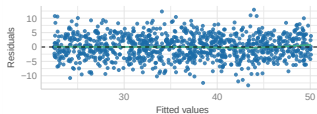
Posterior Predictive Check

Model-predicted lines should resemble observed data line



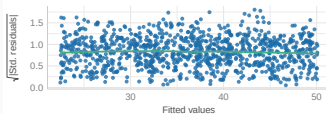
Linearity

Reference line should be flat and horizontal



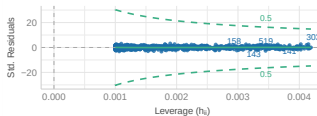
Homogeneity of Variance

Reference line should be flat and horizontal



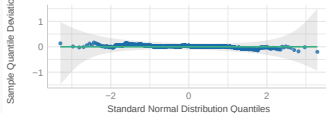
Influential Observations

Points should be inside the contour lines



Normality of Residuals

Dots should fall along the line



A dashboard to explore the full model

```
library("easystats")  
model_dashboard(m1)
```

Using model for prediction

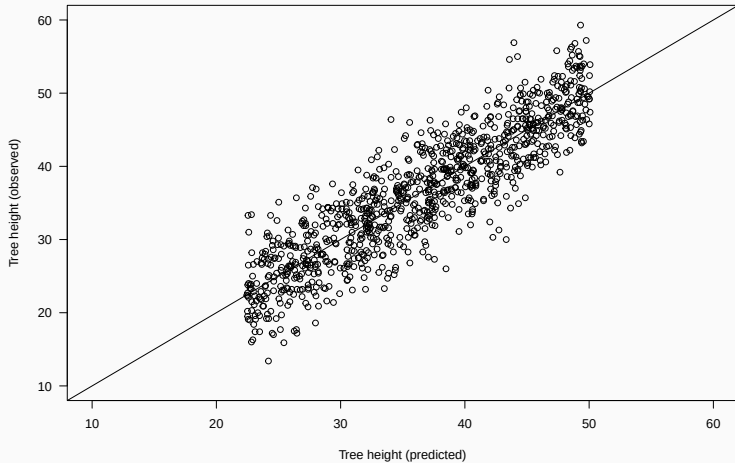
How good is the model in predicting tree height?

`fitted` gives expected value for each observation

```
trees$height.pred <- fitted(m1)
trees$resid <- residuals(m1)
head(trees)
```

	site	dbh	height	sex	dead	height.pred	resid
1	4	29.68	36.1	male	0	37.61328	-1.5132797
2	5	33.29	42.3	male	0	39.83597	2.4640303
3	2	28.03	41.9	female	0	36.59737	5.3026313
4	5	39.86	46.5	female	0	43.88114	2.6188577
5	1	47.94	43.9	female	0	48.85603	-4.9560274
6	1	10.82	26.2	male	0	26.00111	0.1988903

Calibration plot: Observed vs Predicted values



Making predictions for new data

Q: Expected tree height if DBH = 39 cm?

```
new.dbh <- data.frame(dbh = c(39))  
predict(m1, new.dbh, se.fit = TRUE)
```

```
$fit
```

```
1
```

```
43.35164
```

```
$se.fit
```

```
[1] 0.1715514
```

```
$df
```

```
[1] 998
```

```
$residual.scale
```

```
[1] 4.092629
```

Confidence vs Prediction Intervals

Q: Expected tree height if DBH = 39 cm?

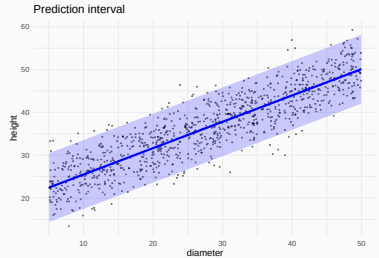
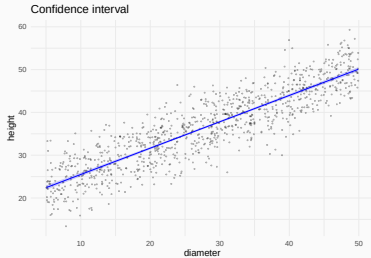
```
predict(m1, new.dbh, interval = "confidence")
```

	fit	lwr	upr
1	43.35164	43.01499	43.68828

```
predict(m1, new.dbh, interval = "prediction")
```

	fit	lwr	upr
1	43.35164	35.31344	51.38983

Confidence vs Prediction Intervals



Making predictions with easystats

Estimate expected values

```
pred <- estimate_expectation(m1)
```

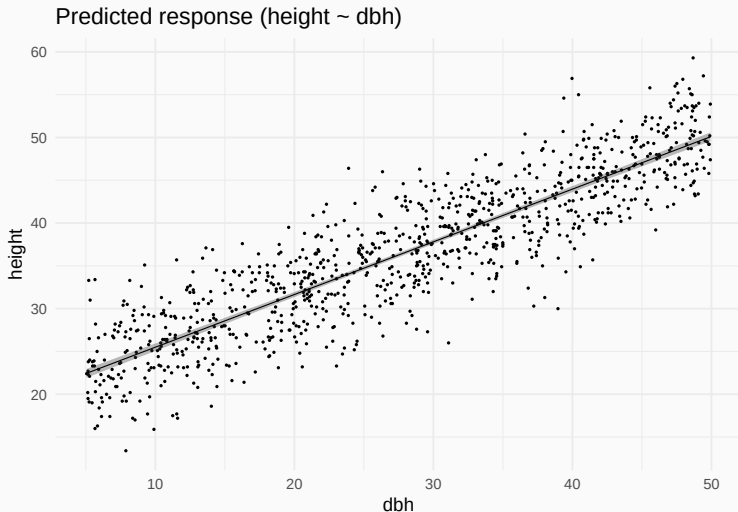
Model-based Expectation

dbh	Predicted	SE	95% CI	Residuals
29.68	37.61	0.13	[37.36, 37.87]	-1.51
33.29	39.84	0.14	[39.56, 40.11]	2.46
28.03	36.60	0.13	[36.34, 36.85]	5.30
39.86	43.88	0.18	[43.53, 44.23]	2.62
47.94	48.86	0.24	[48.38, 49.33]	-4.96
10.82	26.00	0.22	[25.58, 26.42]	0.20

Variable predicted: height

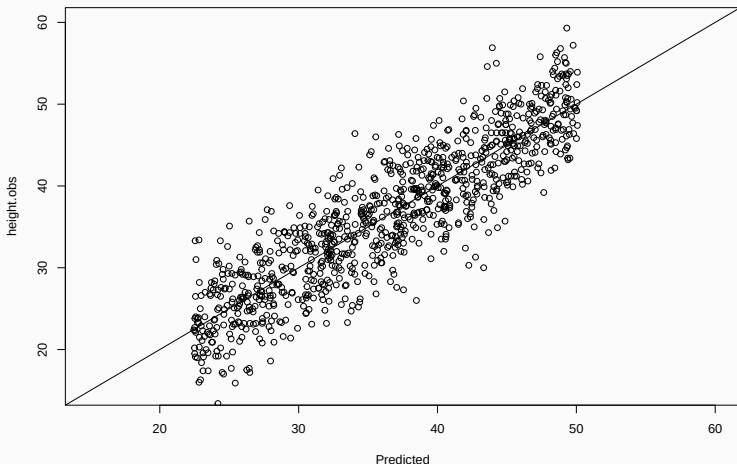
Expected values given DBH

```
plot(estimate_expectation(m1))
```



Calibration plot: observed vs predicted

```
pred$height.obs <- trees$height  
plot(height.obs ~ Predicted, data = pred, xlim = c(15, 60), ylim = c(15, 60))  
abline(a = 0, b = 1)
```



Estimate prediction interval

Accounting for residual variation!

```
pred <- estimate_prediction(m1)
head(pred)
```

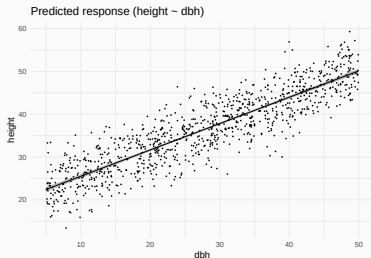
Model-based Prediction

dbh	Predicted	SE	95% CI	Residuals
29.68	37.61	4.09	[29.58, 45.65]	-1.51
33.29	39.84	4.10	[31.80, 47.87]	2.46
28.03	36.60	4.09	[28.56, 44.63]	5.30
39.86	43.88	4.10	[35.84, 51.92]	2.62
47.94	48.86	4.10	[40.81, 56.90]	-4.96
10.82	26.00	4.10	[17.96, 34.04]	0.20

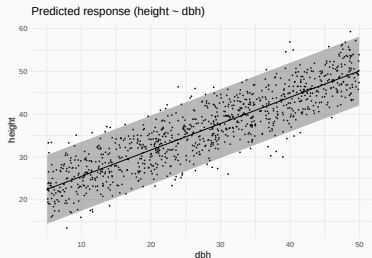
Variable predicted: height

Confidence vs Prediction interval

```
plot(estimate_expectation(m1))
```



```
plot(estimate_prediction(m1))
```



Make predictions for new data

```
estimate_expectation(m1, data = data.frame(dbh = 39))
```

Model-based Expectation

dbh	Predicted	SE	95% CI

39.00	43.35	0.17	[43.01, 43.69]

Variable predicted: height

```
estimate_prediction(m1, data = data.frame(dbh = 39))
```

Model-based Prediction

dbh	Predicted	SE	95% CI

39.00	43.35	4.10	[35.31, 51.39]

```
performance_cv(m1, method = "k_fold", metrics = "common", k = 10)
```

```
# Cross-validation performance (10-fold method)
```

```
RMSE |    R2
```

```
-----
```

```
4.1  | 0.79
```

- Visualise data

- Visualise data
- Understand fitted model (summary)

- Visualise data
- Understand fitted model (summary)
- Visualise model (visreg...)

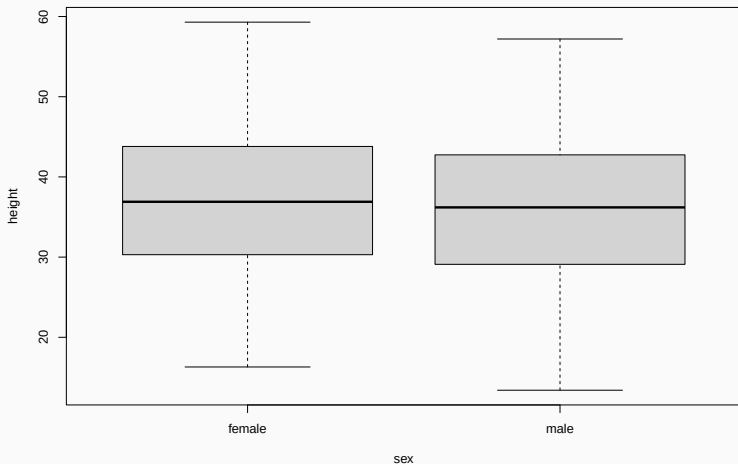
- Visualise data
- Understand fitted model (summary)
- Visualise model (visreg...)
- Check model (plot, check_model, calibration plot...)

- Visualise data
- Understand fitted model (summary)
- Visualise model (visreg...)
- Check model (plot, check_model, calibration plot...)
- Predict (predict, estimate_expectation, estimate_prediction)

Categorical predictors (factors)

Q: Does tree height vary with sex?

```
boxplot(height ~ sex, data = trees)
```



Model height ~ sex

```
m2 <- lm(height ~ sex, data = trees)
```

Call:

```
lm(formula = height ~ sex, data = trees)
```

Residuals:

Min	1Q	Median	3Q	Max
-22.6881	-6.7881	-0.0097	6.7261	22.3687

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	36.9312	0.3981	92.778	<2e-16 ***
sexmale	-0.8432	0.5607	-1.504	0.133

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.865 on 998 degrees of freedom

Multiple R-squared: 0.002261, Adjusted R-squared: 0.001261

F-statistic: 2.261 on 1 and 998 DF, p-value: 0.133

Linear model with categorical predictors

```
m2 <- lm(height ~ sex, data = trees)
```

corresponds to

$$Height_i = a + b_{male} + \varepsilon_i$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

Model height ~ sex

```
m2 <- lm(height ~ sex, data = trees)
```

Call:

```
lm(formula = height ~ sex, data = trees)
```

Residuals:

Min	1Q	Median	3Q	Max
-22.6881	-6.7881	-0.0097	6.7261	22.3687

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	36.9312	0.3981	92.778	<2e-16 ***
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Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.865 on 998 degrees of freedom

Multiple R-squared: 0.002261, Adjusted R-squared: 0.001261

F-statistic: 2.261 on 1 and 998 DF, p-value: 0.133

<https://pollev.com/franciscorod726>

Let's read the model report...

```
report(m2)
```

We fitted a linear model (estimated using OLS) to predict height with sex (formula: `height ~ sex`). The model explains a statistically not significant and very weak proportion of variance ($R^2 = 2.26e-03$, $F(1, 998) = 2.26$, $p = 0.133$, adj. $R^2 = 1.26e-03$). The model's intercept, corresponding to sex = female, is at 36.93 (95% CI [36.15, 37.71], $t(998) = 92.78$, $p < .001$). Within this model:

- The effect of sex [male] is statistically non-significant and negative (beta = -0.84, 95% CI [-1.94, 0.26], $t(998) = -1.50$, $p = 0.133$; Std. beta = -0.10, 95% CI [-0.22, 0.03])

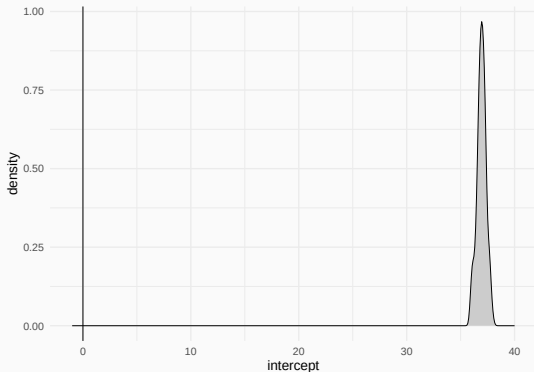
Standardized parameters were obtained by fitting the model on a standardized version of the dataset. 95% Confidence Intervals (CIs) and p-values were computed using a Wald t-distribution approximation.

Estimated distribution of the intercept parameter

Intercept = Height of females

Parameter	Coefficient	SE	95% CI	t(998)	p

(Intercept)	36.93	0.40	[36.15, 37.71]	92.78	< .001

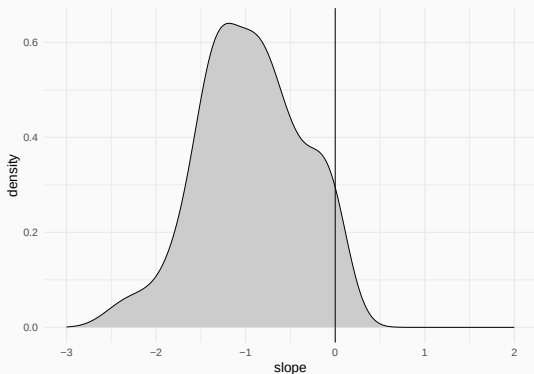


Estimated distribution of the *beta* parameter

beta = height difference of males vs females

Parameter	Coefficient	SE	95% CI	t(998)	p

sex [male]	-0.84	0.56	[-1.94, 0.26]	-1.50	0.133



S-values as alternative to p-values

```
parameters(m2)[2,]
```

Parameter	Coefficient	SE	95% CI	t(998)	p
sex [male]	-0.84	0.56	[-1.94, 0.26]	-1.50	0.133

```
parameters(m2, s_value = TRUE)[2,]
```

Parameter	Coefficient	SE	95% CI	t(998)	s
sex [male]	-0.84	0.56	[-1.94, 0.26]	-1.50	2.91

p-value = 0.133 corresponds to *surprise* of obtaining c. 3 heads in a row when tossing a coin

<https://marginaleffects.com/bonus/svalues.html>

Analysing differences among factor levels

```
library("easystats") # modelbased package  
estimate_means(m2)
```

Estimated Marginal Means

sex	Mean	SE	95% CI
male	36.09	0.39	[35.31, 36.86]
female	36.93	0.40	[36.15, 37.71]

Marginal means estimated at sex

Analysing differences among factor levels

```
estimate_contrasts(m2)
```

Marginal Contrasts Analysis

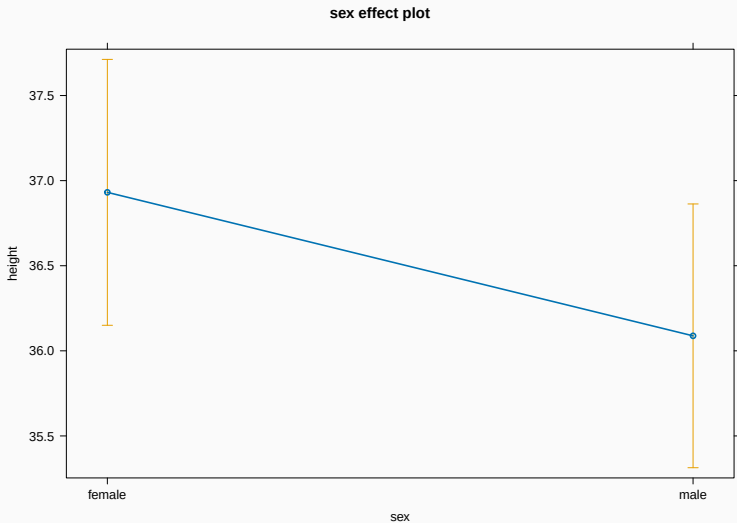
Level1	Level2	Difference	95% CI	SE	t(998)	p
male	female	-0.84	[-1.94, 0.26]	0.56	-1.50	0.133

Marginal contrasts estimated at sex
p-value adjustment method: Holm (1979)

Visualising the fitted model

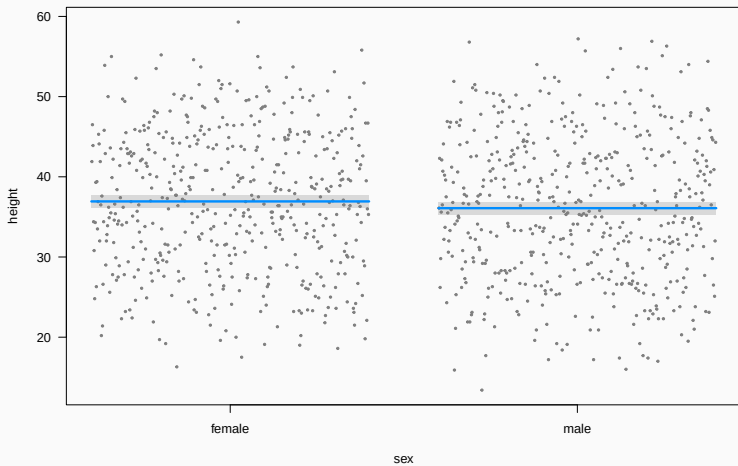
Plot (effects)

```
plot(allEffects(m2))
```

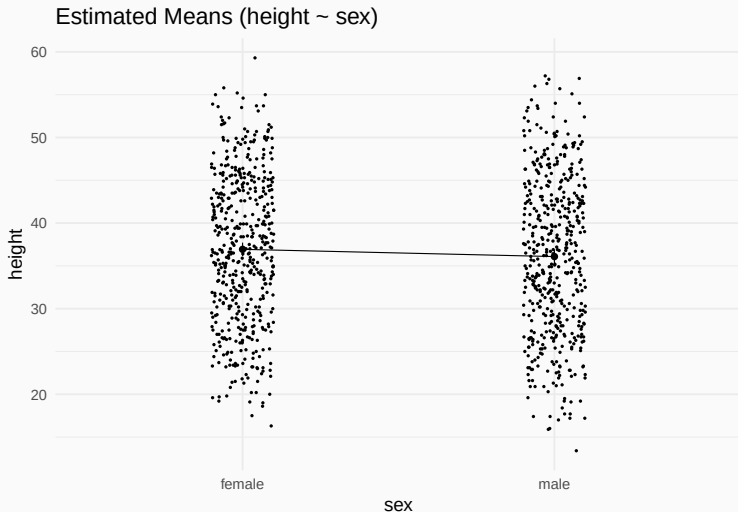


Plot (visreg)

```
visreg(m2)
```

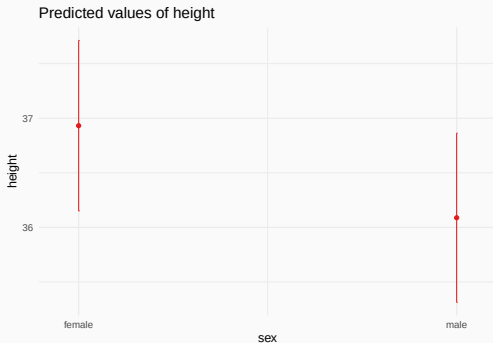


```
plot(estimate_means(m2))
```



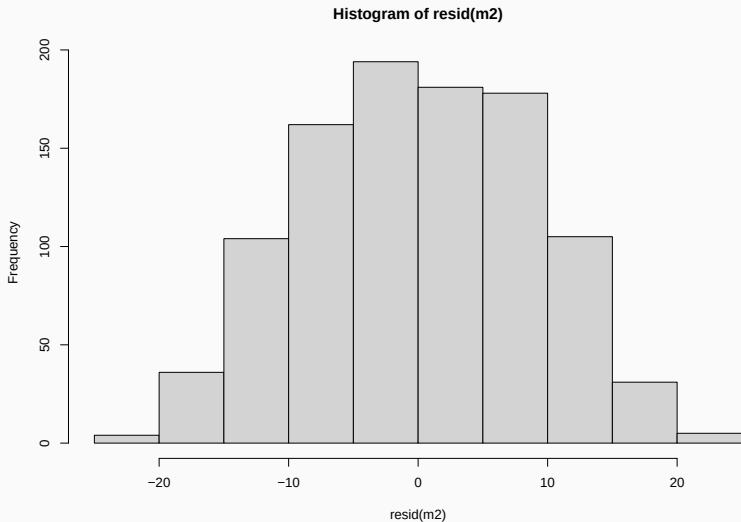
Plot model (sjPlot)

```
library("sjPlot")  
plot_model(m2, type = "eff", terms = "sex")
```

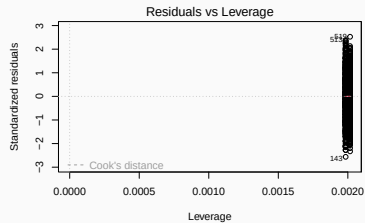
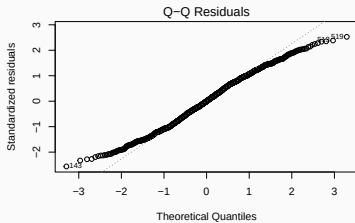
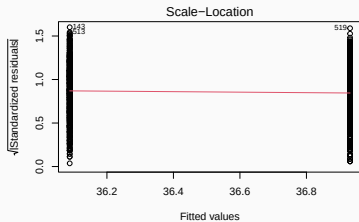
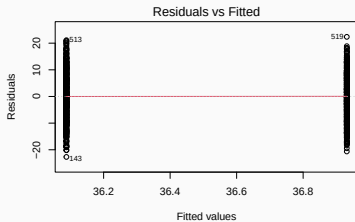


Model checking

```
hist(resid(m2))
```



Model checking: residuals

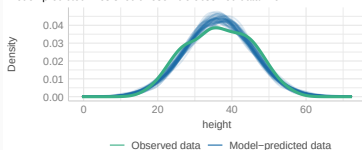


Model checking

```
library("easystats")  
check_model(m2)
```

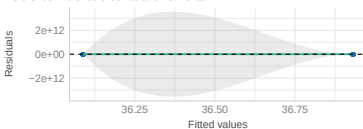
Posterior Predictive Check

Model-predicted lines should resemble observed data line



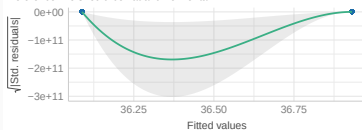
Linearity

Reference line should be flat and horizontal



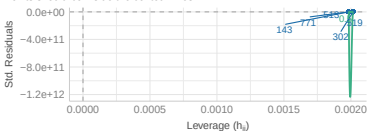
Homogeneity of Variance

Reference line should be flat and horizontal



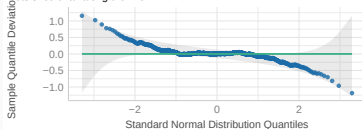
Influential Observations

Points should be inside the contour lines



Normality of Residuals

Points should fall along the line



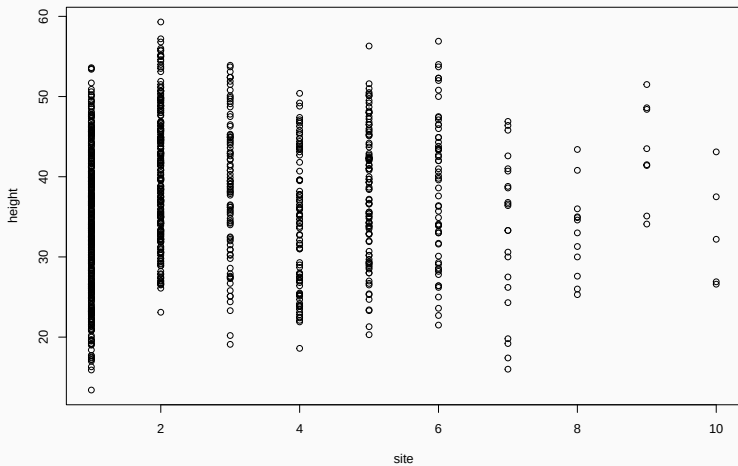
```
model_dashboard(m2)
```

Q: Does height differ among field sites?

<https://pollev.com/franciscorod726>

Plot data first

```
plot(height ~ site, data = trees)
```



Linear model with categorical predictors

```
m3 <- lm(height ~ site, data = trees)
```

$$y_i = a + b_{site2} + c_{site3} + d_{site4} + e_{site5} + \dots + \varepsilon_i$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

Model Height ~ site

All right here?

```
m3 <- lm(height ~ site, data = trees)
```

Call:

```
lm(formula = height ~ site, data = trees)
```

Residuals:

Min	1Q	Median	3Q	Max
-22.4498	-6.7049	0.0709	6.7537	23.0640

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	35.4636	0.4730	74.975	< 2e-16 ***
site	0.3862	0.1413	2.733	0.00639 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.842 on 998 degrees of freedom

Multiple R-squared: 0.007429, Adjusted R-squared: 0.006435

F-statistic: 7.47 on 1 and 998 DF, p-value: 0.006385

Let's check model structure with `equatiomatic`

```
extract_eq(m3)
```

$$\text{height} = \alpha + \beta_1(\text{site}) + \epsilon \quad (3)$$

site is a factor!

```
trees$site <- as.factor(trees$site)
```

Let's check model structure with `equatiomatic`

```
m3 <- lm(height ~ site, data = trees)
extract_eq(m3)
```

$$\text{height} = \alpha + \beta_1(\text{site}_2) + \beta_2(\text{site}_3) + \beta_3(\text{site}_4) + \beta_4(\text{site}_5) + \beta_5(\text{site}_6) + \beta_6(\text{site}_7) + \beta_7(\text{site}_8) \quad (4)$$

Model Height ~ site

Call:

```
lm(formula = height ~ site, data = trees)
```

Residuals:

Min	1Q	Median	3Q	Max
-20.4416	-6.9004	0.0379	6.3051	19.7584

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	33.8416	0.4266	79.329	< 2e-16 ***
site2	6.3411	0.7126	8.899	< 2e-16 ***
site3	4.9991	0.9828	5.086	4.36e-07 ***
site4	0.5329	0.9872	0.540	0.58949
site5	4.3723	0.9425	4.639	3.97e-06 ***
site6	4.7601	1.1709	4.065	5.18e-05 ***
site7	-0.7416	1.8506	-0.401	0.68871
site8	-0.6832	2.4753	-0.276	0.78258
site9	9.1709	3.0165	3.040	0.00243 **
site10	-0.5816	3.8013	-0.153	0.87843

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

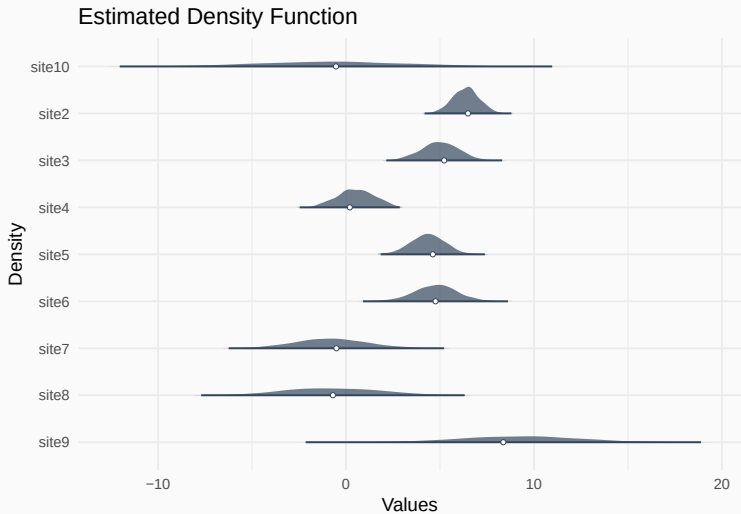
Residual standard error: 8.446 on 990 degrees of freedom

Multiple R-squared: 0.1016, Adjusted R-squared: 0.09344

F-statistic: 12.44 on 9 and 990 DF, p-value: < 2.2e-16

Estimated parameter distributions

```
plot(simulate_parameters(m3), stack = FALSE)
```



Estimated tree heights for each site

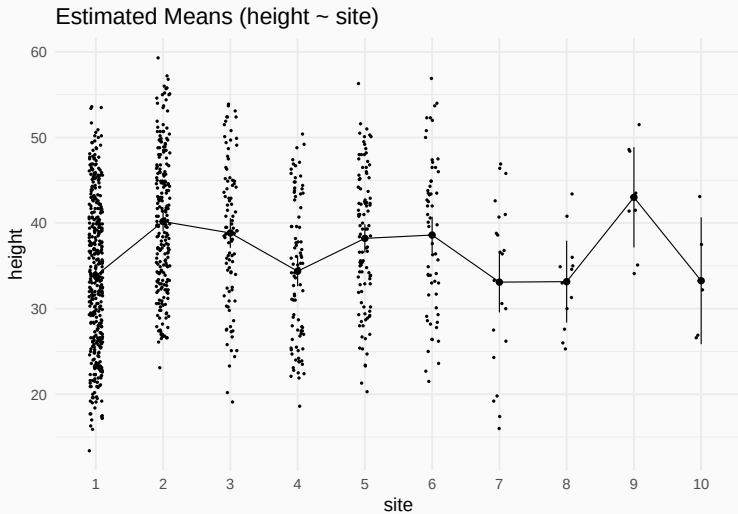
```
estimate_means(m3)
```

Estimated Marginal Means

site	Mean	SE	95% CI
1	33.84	0.43	[33.00, 34.68]
2	40.18	0.57	[39.06, 41.30]
3	38.84	0.89	[37.10, 40.58]
4	34.37	0.89	[32.63, 36.12]
5	38.21	0.84	[36.56, 39.86]
6	38.60	1.09	[36.46, 40.74]
7	33.10	1.80	[29.57, 36.63]
8	33.16	2.44	[28.37, 37.94]
9	43.01	2.99	[37.15, 48.87]
10	33.26	3.78	[25.85, 40.67]

Plot estimated tree heights for each site

```
plot(estimate_means(m3))
```



Analysing differences among factor levels

For finer control see `emmeans` package

```
estimate_contrasts(m3)
```

Marginal Contrasts Analysis

Level1	Level2	Difference	95% CI	SE	t(990)	p
site1	site10	0.58	[-11.85, 13.01]	3.80	0.15	> .999
site1	site2	-6.34	[-8.67, -4.01]	0.71	-8.90	< .001
site1	site3	-5.00	[-8.21, -1.78]	0.98	-5.09	< .001
site1	site4	-0.53	[-3.76, 2.70]	0.99	-0.54	> .999
site1	site5	-4.37	[-7.45, -1.29]	0.94	-4.64	< .001
site1	site6	-4.76	[-8.59, -0.93]	1.17	-4.07	0.002
site1	site7	0.74	[-5.31, 6.79]	1.85	0.40	> .999
site1	site8	0.68	[-7.41, 8.78]	2.48	0.28	> .999
site1	site9	-9.17	[-19.04, 0.69]	3.02	-3.04	0.090
site2	site10	6.92	[-5.57, 19.42]	3.82	1.81	> .999
site2	site3	1.34	[-2.10, 4.79]	1.05	1.27	> .999
site2	site4	5.81	[2.35, 9.27]	1.06	5.49	< .001
site2	site5	1.97	[-1.35, 5.29]	1.02	1.94	> .999
site2	site6	1.58	[-2.44, 5.61]	1.23	1.28	> .999
site2	site7	7.08	[0.90, 13.26]	1.89	3.75	0.008
site2	site8	7.02	[-1.17, 15.21]	2.50	2.81	0.169
site2	site9	-2.83	[-12.77, 7.11]	3.04	-0.93	> .999
site3	site10	5.58	[-7.11, 18.27]	3.88	1.44	> .999
site3	site4	4.47	[0.36, 8.57]	1.26	3.56	0.015
site3	site5	0.63	[-3.37, 4.62]	1.22	0.51	> .999
site3	site6	0.24	[-4.35, 4.83]	1.40	0.17	> .999
site3	site7	5.74	[-0.82, 12.30]	2.01	2.86	0.151
site3	site8	5.68	[-2.80, 14.17]	2.59	2.19	0.084
site3	site9	-4.17	[-14.36, 6.01]	3.11	-1.34	> .999
site4	site10	1.11	[-11.58, 13.81]	3.88	0.29	> .999
site4	site5	-3.84	[-7.84, 0.16]	1.22	-3.14	0.067
site4	site6	-4.23	[-8.83, 0.38]	1.41	-3.00	0.099

Analysing differences among factor levels

How different are site 2 and site 9?

```
library("marginaleffects")  
hypotheses(m3, "site2 = site9")
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
-2.83	3.04	-0.931	0.352	1.5	-8.79	3.13

Term: site2 = site9

Columns: term, estimate, std.error, statistic, p.value, s.value, co

Presenting model results

```
parameters(m3)
```

Parameter	Coefficient	SE	95% CI	t(990)	p
(Intercept)	33.84	0.43	[33.00, 34.68]	79.33	< .001
site [2]	6.34	0.71	[4.94, 7.74]	8.90	< .001
site [3]	5.00	0.98	[3.07, 6.93]	5.09	< .001
site [4]	0.53	0.99	[-1.40, 2.47]	0.54	0.589
site [5]	4.37	0.94	[2.52, 6.22]	4.64	< .001
site [6]	4.76	1.17	[2.46, 7.06]	4.07	< .001
site [7]	-0.74	1.85	[-4.37, 2.89]	-0.40	0.689
site [8]	-0.68	2.48	[-5.54, 4.17]	-0.28	0.783
site [9]	9.17	3.02	[3.25, 15.09]	3.04	0.002
site [10]	-0.58	3.80	[-8.04, 6.88]	-0.15	0.878

Presenting model results

```
estimate_means(m3)
```

Estimated Marginal Means

site	Mean	SE	95% CI
1	33.84	0.43	[33.00, 34.68]
2	40.18	0.57	[39.06, 41.30]
3	38.84	0.89	[37.10, 40.58]
4	34.37	0.89	[32.63, 36.12]
5	38.21	0.84	[36.56, 39.86]
6	38.60	1.09	[36.46, 40.74]
7	33.10	1.80	[29.57, 36.63]
8	33.16	2.44	[28.37, 37.94]
9	43.01	2.99	[37.15, 48.87]
10	33.26	3.78	[25.85, 40.67]

Marginal means estimated at site

Presenting model results

```
modelsummary(m3, estimate = "{estimate} ({std.error})", statistic = NULL,  
  fmt = 1, gof_map = NA, coef_rename = paste0("site", 1:10), output = "markdown")
```

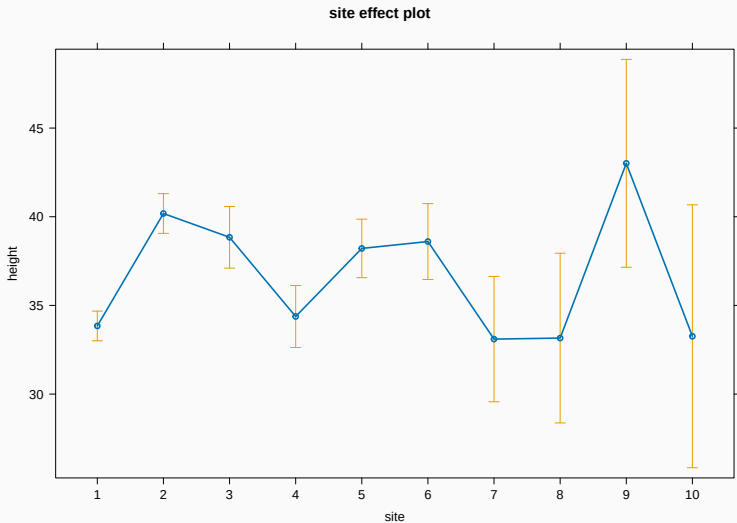
	(1)
site1	33.8 (0.4)
site2	6.3 (0.7)
site3	5.0 (1.0)
site4	0.5 (1.0)
site5	4.4 (0.9)
site6	4.8 (1.2)
site7	-0.7 (1.9)
site8	-0.7 (2.5)
site9	9.2 (3.0)
site10	-0.6 (3.8)

Presenting model results

```
library("gtsummary")  
tbl_regression(m3)
```

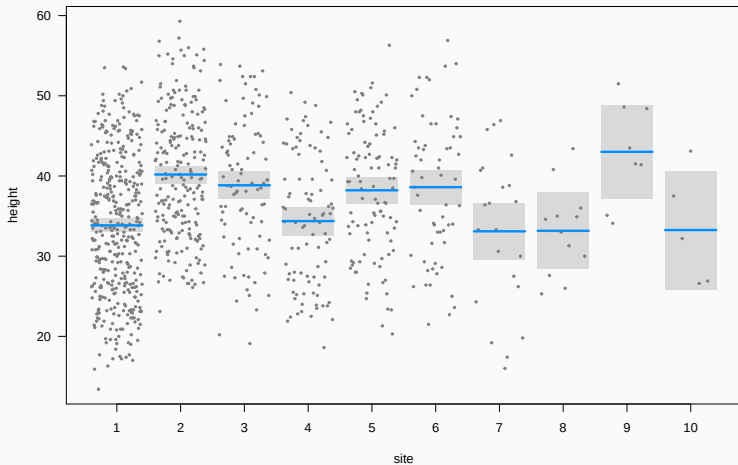
Characteristic	Beta	95% CI ¹	p-value
site			
1	—	—	
2	6.3	4.9, 7.7	<0.001
3	5.0	3.1, 6.9	<0.001
4	0.53	-1.4, 2.5	0.6
5	4.4	2.5, 6.2	<0.001
6	4.8	2.5, 7.1	<0.001
7	-0.74	-4.4, 2.9	0.7
8	-0.68	-5.5, 4.2	0.8
9	9.2	3.3, 15	0.002


```
plot(allEffects(m3))
```

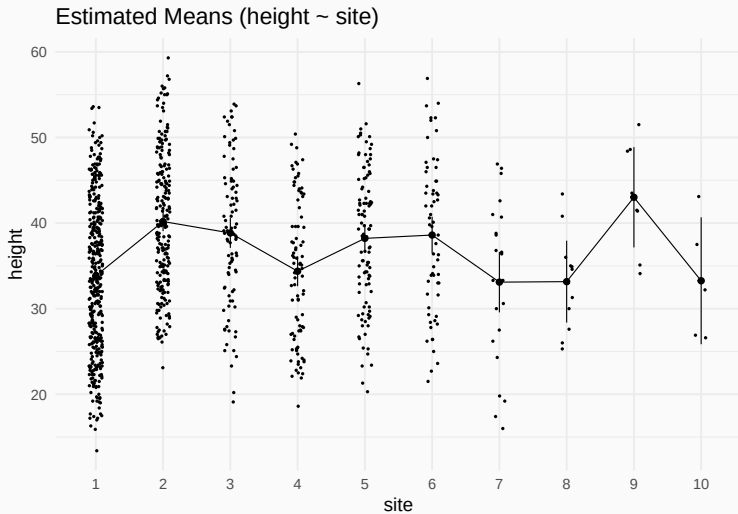


Plot (visreg)

```
visreg(m3)
```

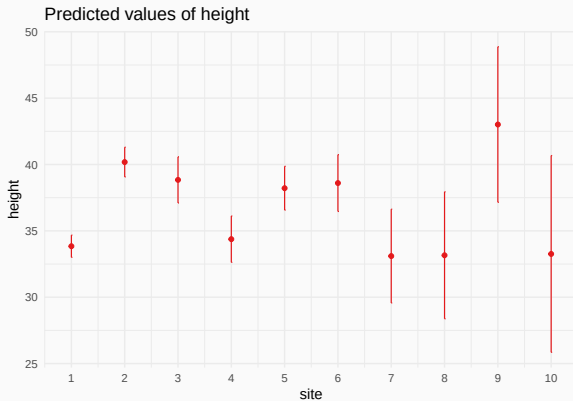


```
plot(estimate_means(m3))
```



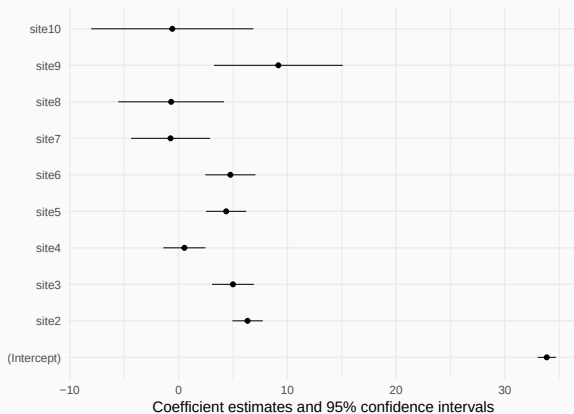
Plot model (sjPlot)

```
plot_model(m3, type = "eff", terms = "site")
```



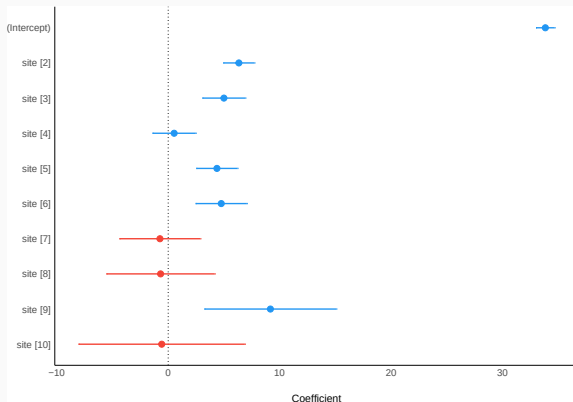
Plot model (modelsummary)

```
modelplot(m3)
```



Plot model (easystats)

```
plot(parameters(m3), show_intercept = TRUE)
```



Fit model without intercept

```
m3bis <- lm(height ~ site - 1, data = trees)
```

Call:

```
lm(formula = height ~ site - 1, data = trees)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-20.4416	-6.9004	0.0379	6.3051	19.7584

Coefficients:

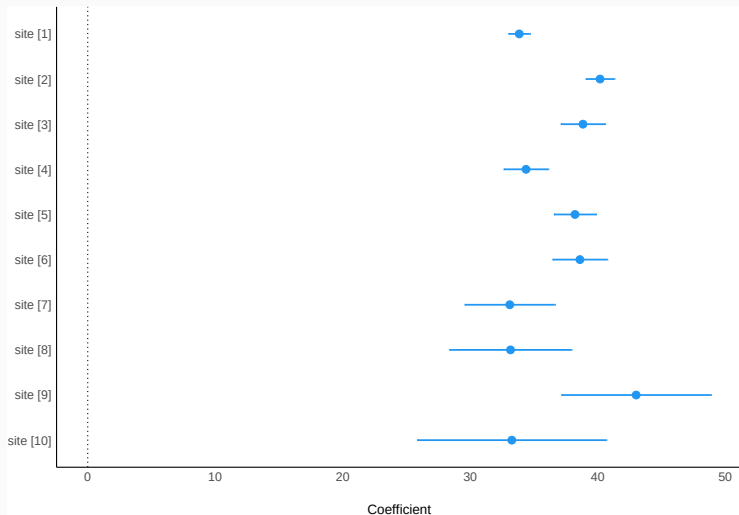
	Estimate	Std. Error	t value	Pr(> t)
site1	33.8416	0.4266	79.329	<2e-16 ***
site2	40.1826	0.5707	70.404	<2e-16 ***
site3	38.8407	0.8854	43.868	<2e-16 ***
site4	34.3744	0.8903	38.610	<2e-16 ***
site5	38.2139	0.8404	45.469	<2e-16 ***
site6	38.6017	1.0904	35.401	<2e-16 ***
site7	33.1000	1.8007	18.381	<2e-16 ***
site8	33.1583	2.4382	13.599	<2e-16 ***
site9	43.0125	2.9862	14.404	<2e-16 ***
site10	33.2600	3.7773	8.805	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

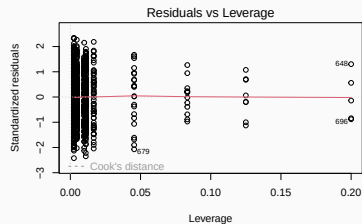
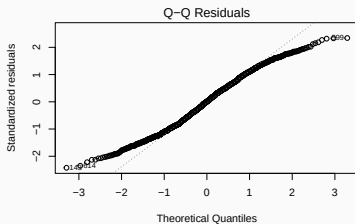
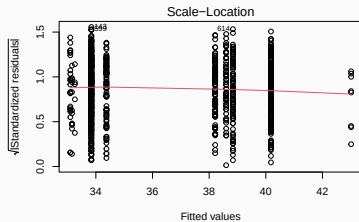
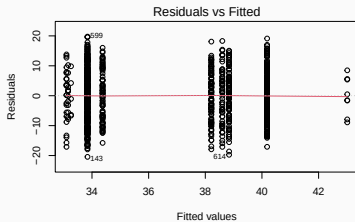
Residual standard error: 8.446 on 990 degrees of freedom

Model without intercept

```
plot(parameters(m3bis))
```



Model checking: residuals

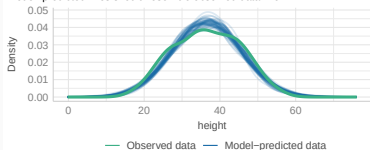


Model checking: residuals

`check_model(m3)`

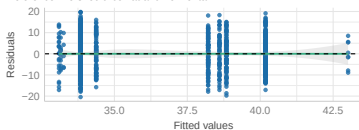
Posterior Predictive Check

Model-predicted lines should resemble observed data line



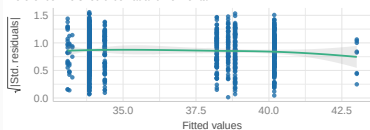
Linearity

Reference line should be flat and horizontal



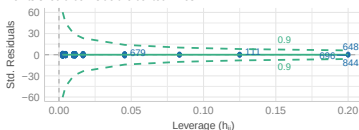
Homogeneity of Variance

Reference line should be flat and horizontal



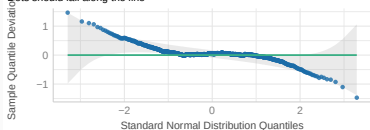
Influential Observations

Points should be inside the contour lines



Normality of Residuals

Points should fall along the line



Combining continuous and categorical predictors

Predicting tree height based on dbh and site

```
lm(height ~ site + dbh, data = trees)
```

corresponds to

$$y_i = a + b_{site2} + c_{site3} + d_{site4} + e_{site5} + \dots + k \cdot DBH_i + \varepsilon_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

Predicting tree height based on dbh and site

Call:

```
lm(formula = height ~ site + dbh, data = trees)
```

Residuals:

Min	1Q	Median	3Q	Max
-10.1130	-1.9885	0.0582	2.0314	11.3320

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	16.699037	0.260565	64.088	< 2e-16 ***
site2	6.504303	0.256730	25.335	< 2e-16 ***
site3	4.357457	0.354181	12.303	< 2e-16 ***
site4	1.934650	0.356102	5.433	6.98e-08 ***
site5	3.637432	0.339688	10.708	< 2e-16 ***
site6	4.204511	0.421906	9.966	< 2e-16 ***
site7	-0.176193	0.666772	-0.264	0.7916
site8	-5.312648	0.893603	-5.945	3.82e-09 ***
site9	5.437049	1.087766	4.998	6.84e-07 ***
site10	2.263338	1.369986	1.652	0.0988 .
dbh	0.617075	0.007574	81.473	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.043 on 989 degrees of freedom

Multiple R-squared: 0.8835, Adjusted R-squared: 0.8823

Presenting model results

```
parameters(m4)
```

Parameter	Coefficient	SE	95% CI	t(989)	p
(Intercept)	16.70	0.26	[16.19, 17.21]	64.09	< .001
site [2]	6.50	0.26	[6.00, 7.01]	25.34	< .001
site [3]	4.36	0.35	[3.66, 5.05]	12.30	< .001
site [4]	1.93	0.36	[1.24, 2.63]	5.43	< .001
site [5]	3.64	0.34	[2.97, 4.30]	10.71	< .001
site [6]	4.20	0.42	[3.38, 5.03]	9.97	< .001
site [7]	-0.18	0.67	[-1.48, 1.13]	-0.26	0.792
site [8]	-5.31	0.89	[-7.07, -3.56]	-5.95	< .001
site [9]	5.44	1.09	[3.30, 7.57]	5.00	< .001
site [10]	2.26	1.37	[-0.43, 4.95]	1.65	0.099
dbh	0.62	7.57e-03	[0.60, 0.63]	81.47	< .001

Estimated tree heights for each site

```
estimate_means(m4)
```

Estimated Marginal Means

site	Mean	SE	95% CI
1	33.90	0.15	[33.60, 34.21]
2	40.41	0.21	[40.01, 40.81]
3	38.26	0.32	[37.64, 38.89]
4	35.84	0.32	[35.21, 36.47]
5	37.54	0.30	[36.95, 38.14]
6	38.11	0.39	[37.34, 38.88]
7	33.73	0.65	[32.45, 35.00]
8	28.59	0.88	[26.86, 30.32]
9	39.34	1.08	[37.23, 41.45]
10	36.17	1.36	[33.50, 38.84]

Fit model without intercept

```
m4 <- lm(height ~ -1 + site + dbh, data = trees)
```

Call:

```
lm(formula = height ~ -1 + site + dbh, data = trees)
```

Residuals:

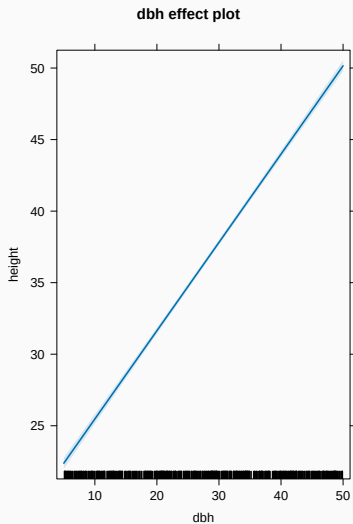
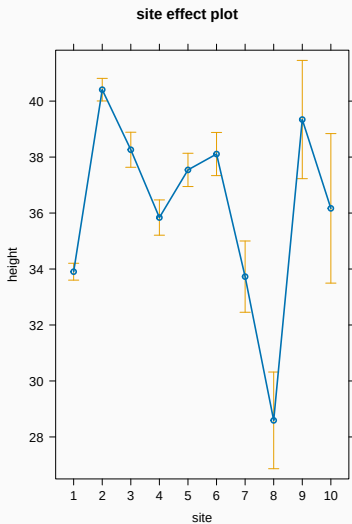
	Min	1Q	Median	3Q	Max
	-10.1130	-1.9885	0.0582	2.0314	11.3320

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
site1	16.699037	0.260565	64.09	<2e-16 ***
site2	23.203340	0.292773	79.25	<2e-16 ***
site3	21.056494	0.386532	54.48	<2e-16 ***
site4	18.633687	0.374456	49.76	<2e-16 ***
site5	20.336469	0.373942	54.38	<2e-16 ***
site6	20.903548	0.448913	46.56	<2e-16 ***
site7	16.522844	0.679936	24.30	<2e-16 ***
site8	11.386389	0.918198	12.40	<2e-16 ***
site9	22.136086	1.105970	20.02	<2e-16 ***
site10	18.962375	1.372158	13.82	<2e-16 ***
dbh	0.617075	0.007574	81.47	<2e-16 ***

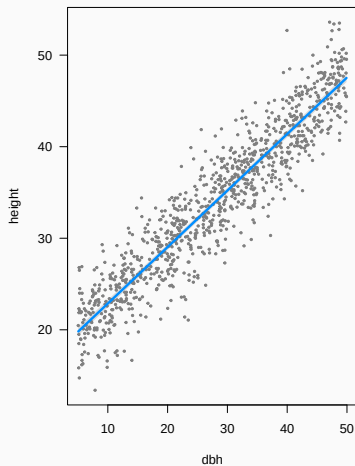
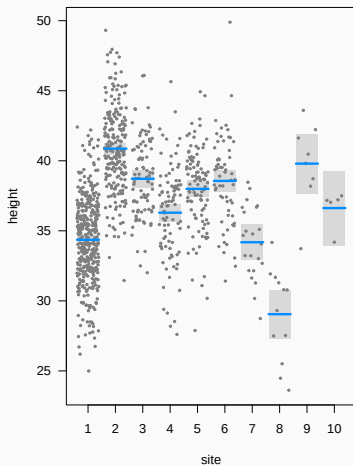
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


```
plot(allEffects(m4))
```



Plot (visreg)

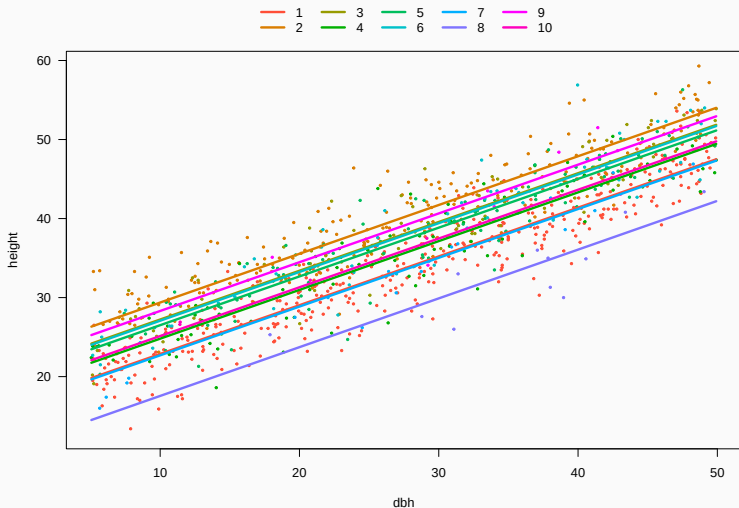
```
visreg(m4)
```



```
null device
```

Plot (visreg)

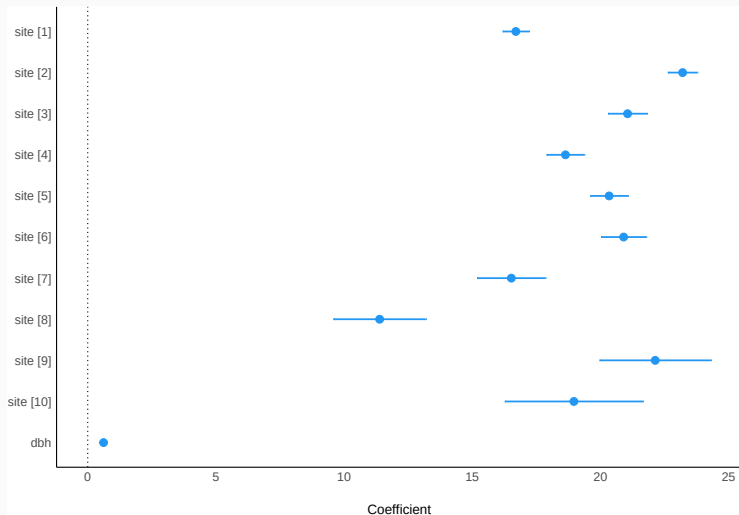
```
visreg(m4, xvar = "dbh", by = "site", overlay = TRUE, band = FALSE)
```



```
plot_model(m4, type = "eff", terms = "site")
```

Plot model (easystats)

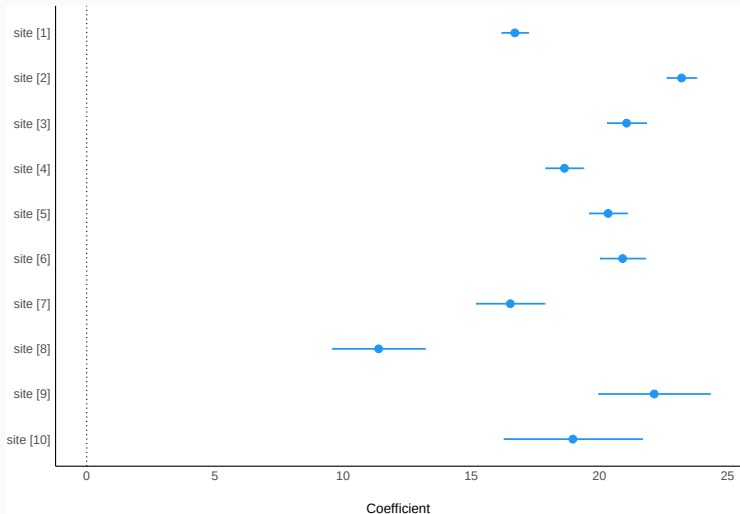
```
plot(parameters(m4))
```



Plot model (easystats)

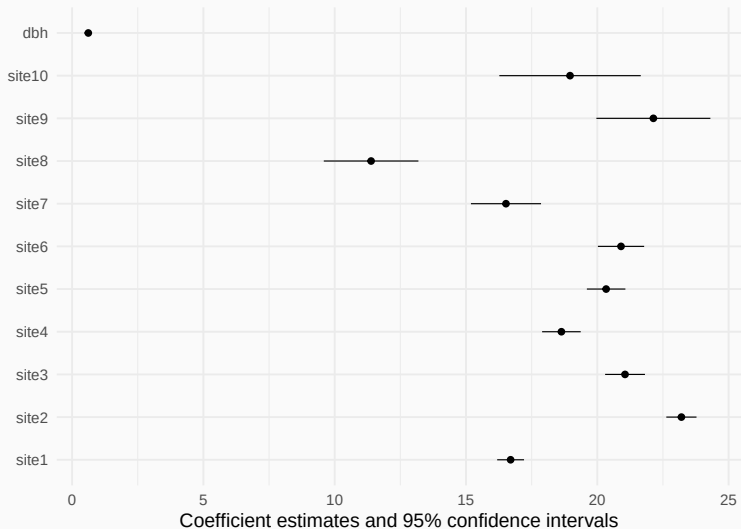
Keeping sites only, dropping "dbh"

```
plot(parameters(m4, drop = "dbh"))
```



Plot model (modelsummary)

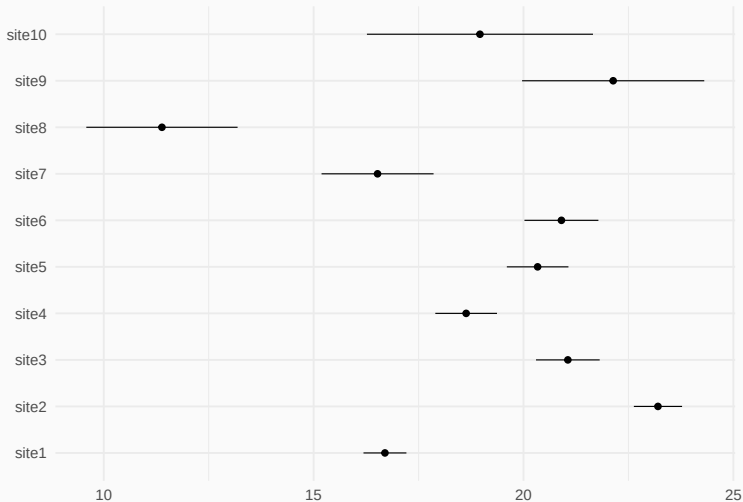
```
modelplot(m4)
```



Plot model (modelsummary)

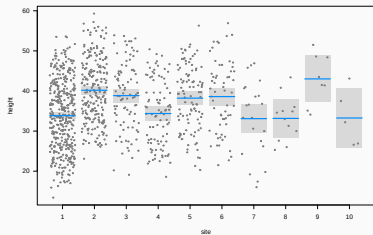
Keeping sites only, dropping "dbh"

```
modelplot(m4, coef_omit = "dbh")
```

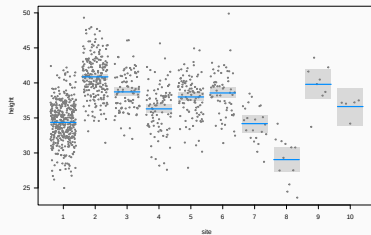


What happened to site 8?

```
visreg(m3)
```

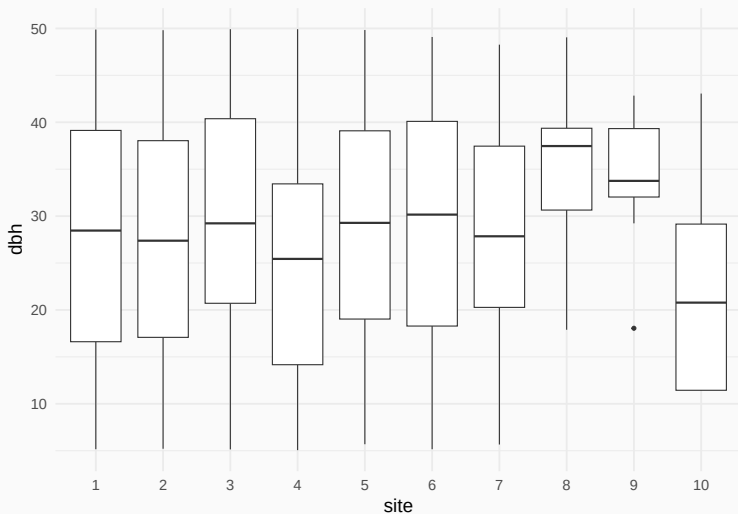


```
visreg(m4, xvar = "site")
```



What happened to site 8?

site 8 has the largest diameters



What happened to site 8?

DBH

```
aggregate(trees$dbh ~ trees$site, FUN = me
```

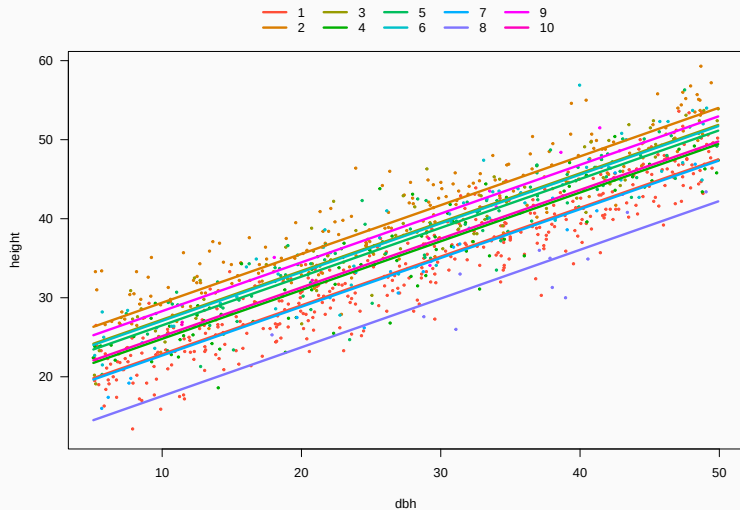
	trees\$site	trees\$dbh
1	1	27.78033
2	2	27.51580
3	3	28.82011
4	4	25.50867
5	5	28.97119
6	6	28.68067
7	7	26.86409
8	8	35.28250
9	9	33.83125
10	10	23.17000

HEIGHT

```
aggregate(trees$height ~ trees$site, FUN =
```

	trees\$site	trees\$height
1	1	33.84158
2	2	40.18265
3	3	38.84066
4	4	34.37444
5	5	38.21386
6	6	38.60167
7	7	33.10000
8	8	33.15833
9	9	43.01250
10	10	33.26000

We have fitted model w/ many intercepts and single slope

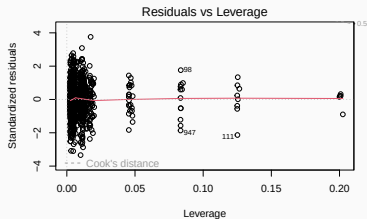
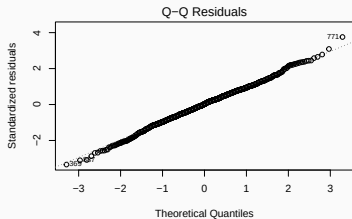
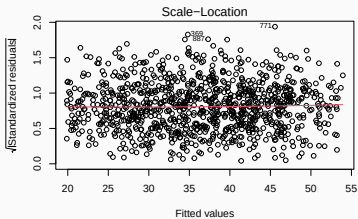
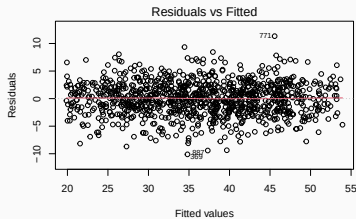


Slope is the same for all sites

```
parameters(m4, keep = "dbh")
```

Parameter	Coefficient	SE	95% CI	t(989)	p
dbh	0.62	7.57e-03	[0.60, 0.63]	81.47	< .001

Model checking: residuals

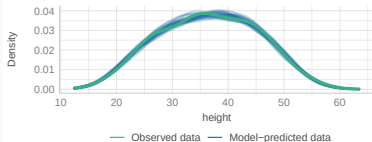


Model checking: residuals

`check_model(m4)`

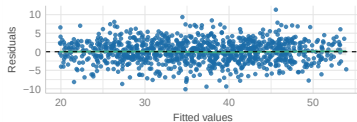
Posterior Predictive Check

Model-predicted lines should resemble observed data line



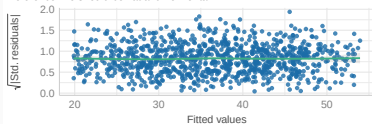
Linearity

Reference line should be flat and horizontal



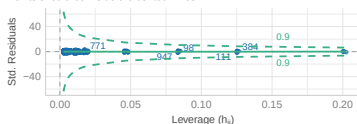
Homogeneity of Variance

Reference line should be flat and horizontal



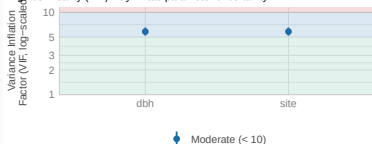
Influential Observations

Points should be inside the contour lines



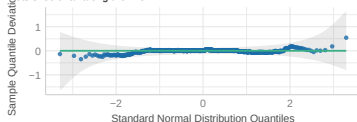
Collinearity

High collinearity (VIF) may inflate parameter uncertainty



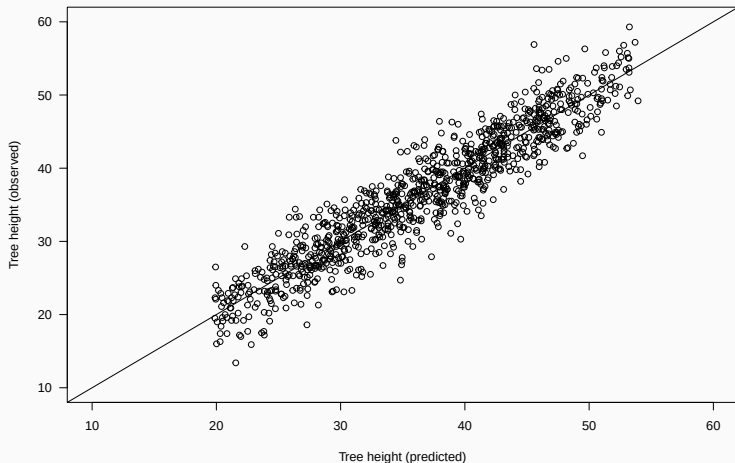
Normality of Residuals

Dots should fall along the line



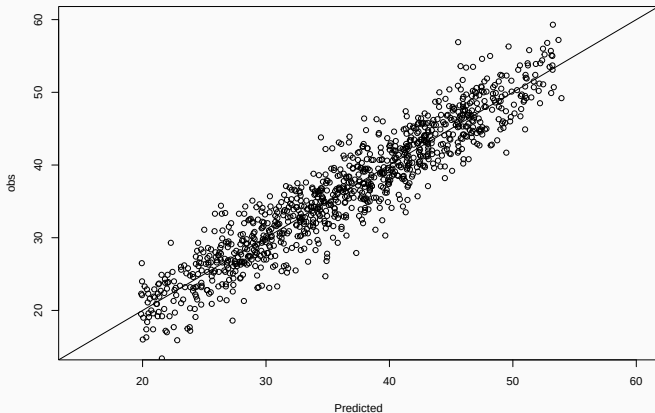
How good is this model? Calibration plot

```
trees$height.pred <- fitted(m4)
plot(trees$height.pred, trees$height, xlab = "Tree height (predicted)",
     ylab = "Tree height (observed)", abline(a = 0, b = 1))
```



How good is this model? Calibration plot (easystats)

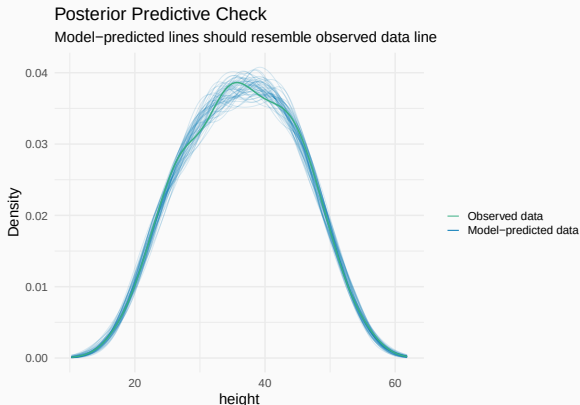
```
pred <- estimate_expectation(m4)
pred$obs <- trees$height
plot(obs ~ Predicted, data = pred, xlim = c(15, 60), ylim = c(15, 60))
abline(a = 0, b = 1)
```



Posterior predictive checking

Simulating response data from fitted model (y_{rep})
and comparing with observed response (y)

```
performance::check_predictions(m4)
```



Predicting heights of new trees

Using model for prediction

Expected height of 10-cm diameter tree in each site?

```
trees.10cm <- data.frame(site = as.factor(1:10),  
                           dbh = 10)  
trees.10cm
```

	site	dbh
1	1	10
2	2	10
3	3	10
4	4	10
5	5	10
6	6	10
7	7	10
8	8	10
9	9	10
10	10	10

Using model for prediction

Confidence interval

```
predict(m4, newdata = trees.10cm, interval = "confidence")
```

	fit	lwr	upr
1	22.86979	22.46878	23.27079
2	29.37409	28.89388	29.85430
3	27.22724	26.54160	27.91289
4	24.80444	24.13410	25.47477
5	26.50722	25.84952	27.16492
6	27.07430	26.25490	27.89370
7	22.69359	21.39601	23.99117
8	17.55714	15.79282	19.32146
9	28.30683	26.16606	30.44761
10	25.13312	22.45540	27.81085

Using model for prediction

Prediction interval (accounting for residual variance)

```
predict(m4, newdata = trees.10cm, interval = "prediction")
```

	fit	lwr	upr
1	22.86979	16.88478	28.85480
2	29.37409	23.38325	35.36493
3	27.22724	21.21645	33.23804
4	24.80444	18.79537	30.81350
5	26.50722	20.49955	32.51489
6	27.07430	21.04678	33.10181
7	22.69359	16.58268	28.80451
8	17.55714	11.33039	23.78388
9	28.30683	21.96314	34.65053
10	25.13312	18.58868	31.67757

Using model for prediction

Prediction interval (99%)

```
predict(m4, newdata = trees.10cm, interval = "prediction",  
        level = 0.99)
```

	fit	lwr	upr
1	22.86979	14.998587	30.74098
2	29.37409	21.495225	37.25295
3	27.22724	19.322133	35.13235
4	24.80444	16.901598	32.70727
5	26.50722	18.606216	34.40822
6	27.07430	19.147195	35.00140
7	22.69359	14.656813	30.73037
8	17.55714	9.368019	25.74626
9	28.30683	19.963913	36.64976
10	25.13312	16.526183	33.74007

Predicting heights of new trees (easystats)

Using model for prediction

Expected height of 10-cm diameter tree in each site?

```
trees.10cm <- data.frame(site = as.factor(1:10),  
                           dbh = 10)  
trees.10cm
```

	site	dbh
1	1	10
2	2	10
3	3	10
4	4	10
5	5	10
6	6	10
7	7	10
8	8	10
9	9	10
10	10	10

Using model for prediction

Expected height of 10-cm DBH trees at each site

```
pred <- estimate_expectation(m4, data = trees.10cm)
```

Model-based Expectation

site	dbh	Predicted	SE	95% CI
1	10.00	22.87	0.20	[22.47, 23.27]
2	10.00	29.37	0.24	[28.89, 29.85]
3	10.00	27.23	0.35	[26.54, 27.91]
4	10.00	24.80	0.34	[24.13, 25.47]
5	10.00	26.51	0.34	[25.85, 27.16]
6	10.00	27.07	0.42	[26.25, 27.89]
7	10.00	22.69	0.66	[21.40, 23.99]
8	10.00	17.56	0.90	[15.79, 19.32]
9	10.00	28.31	1.09	[26.17, 30.45]
10	10.00	25.13	1.36	[22.46, 27.81]

Variable predicted: height

Using model for prediction

Prediction intervals (accounting for residual variance)

```
pred <- estimate_prediction(m4, data = trees.10cm)
```

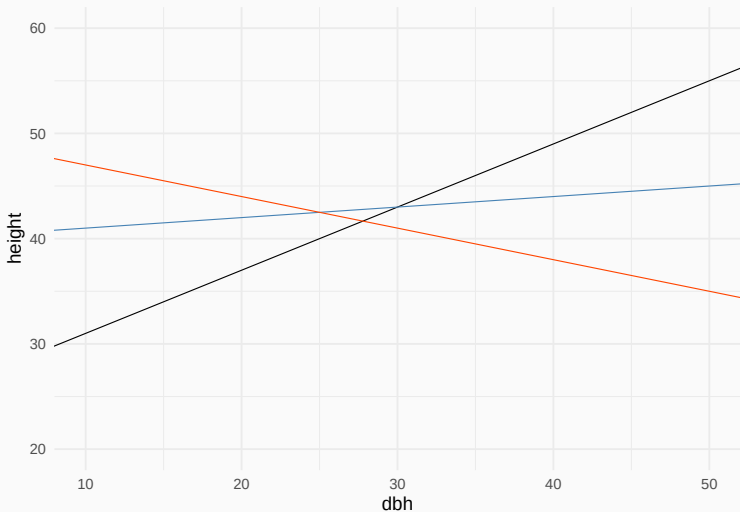
Model-based Prediction

site	dbh	Predicted	SE	95% CI
1	10.00	22.87	3.05	[16.88, 28.85]
2	10.00	29.37	3.05	[23.38, 35.36]
3	10.00	27.23	3.06	[21.22, 33.24]
4	10.00	24.80	3.06	[18.80, 30.81]
5	10.00	26.51	3.06	[20.50, 32.51]
6	10.00	27.07	3.07	[21.05, 33.10]
7	10.00	22.69	3.11	[16.58, 28.80]
8	10.00	17.56	3.17	[11.33, 23.78]
9	10.00	28.31	3.23	[21.96, 34.65]
10	10.00	25.13	3.33	[18.59, 31.68]

Variable predicted: height

Q: Does allometric relationship
between Height and Diameter
vary among sites?

Does allometric relationship between Height and Diameter vary among sites?



Model with interactions

Call:

```
lm(formula = height ~ site * dbh, data = trees)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-10.1017	-1.9839	0.0645	2.0486	11.1789

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	16.359437	0.360054	45.436	< 2e-16 ***
site2	7.684781	0.609657	12.605	< 2e-16 ***
site3	4.518568	0.867008	5.212	2.28e-07 ***
site4	2.769336	0.813259	3.405	0.000688 ***
site5	3.917607	0.870983	4.498	7.68e-06 ***
site6	4.155161	1.009379	4.117	4.17e-05 ***
site7	-2.306799	1.551303	-1.487	0.137334
site8	-2.616095	4.090671	-0.640	0.522630
site9	2.621560	5.073794	0.517	0.605492
site10	4.662340	2.991072	1.559	0.119378
dbh	0.629299	0.011722	53.685	< 2e-16 ***
site2:dbh	-0.042784	0.020033	-2.136	0.032950 *
site3:dbh	-0.006031	0.027640	-0.218	0.827312
site4:dbh	-0.031633	0.028225	-1.121	0.262677
site5:dbh	-0.010173	0.027887	-0.365	0.715334
site6:dbh	0.001337	0.032109	0.042	0.966797
site7:dbh	0.079728	0.052056	1.532	0.125951
site8:dbh	-0.079027	0.113386	-0.697	0.485984
site9:dbh	0.081035	0.146649	0.553	0.580679
site10:dbh	-0.101107	0.114520	-0.883	0.377522

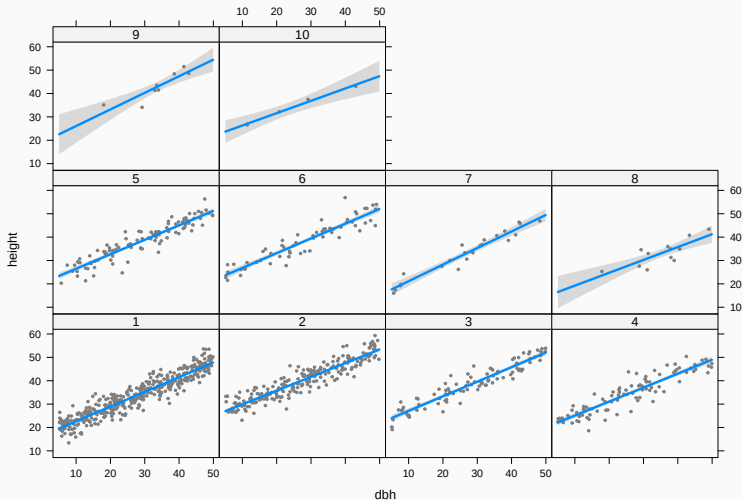
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.041 on 980 degrees of freedom

Multiple R-squared: 0.8847 Adjusted R-squared: 0.8825

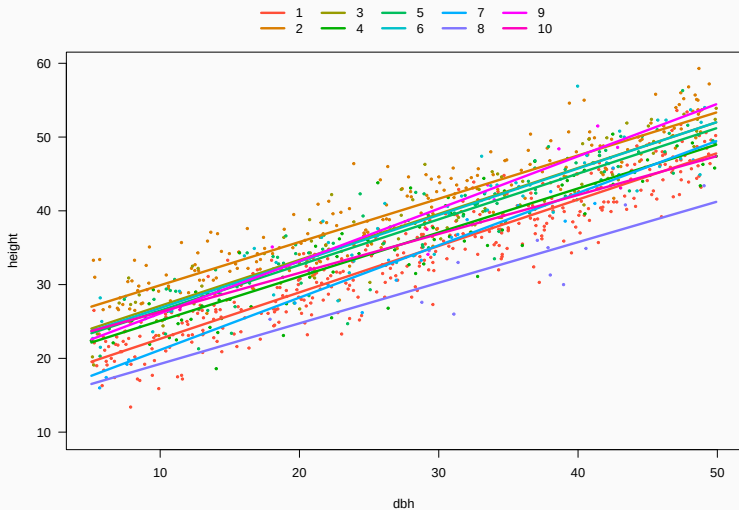
Does slope vary among sites?

```
visreg(m5, xvar = "dbh", by = "site")
```



Does slope vary among sites?

```
visreg(m5, xvar = "dbh", by = "site", overlay = TRUE, band = FALSE)
```



Does slope vary among sites?

```
library("marginaleffects")  
hypotheses(m5, "`site9:dbh` = `site10:dbh`")
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
0.182	0.185	0.983	0.326	1.6	-0.181	0.545

Term: 'site9:dbh' = 'site10:dbh'

Columns: term, estimate, std.error, statistic, p.value, s.value, conf.low, conf.high

Examining fitted model with {modelStudio}

```
library("modelStudio")  
m5.explain <- DALEX::explain(  
  m5,  
  data = trees,  
  y = trees$height)  
modelStudio(m5.explain, viewer = "browser")
```

- [paperplanes](#): How does flight distance differ with age, gender or paper type?

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- [racing pigeons](#): is speed related to sex?